Introduction to Probability

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Preface

Embarking on the study of probability is both an opportunity and a challenge. This course represents a critical foundation for all subsequent studies in econometrics and data science. As we navigate an era of big data and artificial intelligence, proficiency in probability theory is indispensable. It not only builds the foundations for higher level courses but also cultivates a mindset geared towards probabilistic reasoning, which is essential for mitigating cognitive biases and making more informed decisions. Human cognition is often not well-equipped to handle probabilistic thinking. As we will see, human intuition and heuristics are mostly wrong about probabilistic events. This course seeks to provide a mathematical framework for properly understanding and applying probability.

Probability theory is more than just a mathematical discipline; it is a vital tool for making sense of uncertainty in the real world. Consider the myriad of questions that we encounter daily: Will it rain tomorrow? What is the expected return on an investment? What are the odds of a particular political party winning an election? How can a business optimize its customer service strategy when customer arrival times are unpredictable? Probability theory provides a scientific approach to answering these questions, enabling us to model and analyze uncertainties with mathematical tools.

However, this journey is not without its difficulties. For freshmen, particularly those new to calculus and linear algebra, the course presents a steep learning curve. The breadth of new concepts—such as random variables, expectations, and various distributions—can be overwhelming if encountered for the first time. Additionally, the use of advanced calculus, particularly integrals, may pose challenges for those who are still familiarizing themselves with these mathematical tools.

Despite these challenges, the rewards of studying probability are substantial.

Gaining a deep understanding of probability will not only enhance your knowledge base but also fundamentally transform your approach to problem-solving. The principles you will learn are applicable to a wide range of fields beyond econometrics and data science, including engineering, finance, and social sciences. You will learn the tools to approach these problems systematically and make informed decisions based on statistical evidence and probability. So be prepared for a challenging and rewarding journey!

Learning objectives:

- **Review fundamental concepts:** Revisit the probability and calculus concepts learned in high school to ensure a solid foundation for more advanced topics.
- **Understand core probability theory:** Gain a thorough understanding of key concepts and theorems in probability theory, including random variables, expectations, covariances, and so on.
- **Develop probabilistic thinking:** Learn to approach problems with a probabilistic mindset and use random variables to describe and analyze uncertain outcomes.
- **Model real-world events:** Identify and apply important probability distributions to model and interpret real-world phenomena effectively.
- **Enjoy and have fun:** Discover and appreciate the inherent elegance of mathematics and the beauty of probability theory.

Study tips for new college students:

- **Limit electronic distractions:** While digital tools like slides and tablets are convenient, traditional paper and pencil methods remain the most effective way to engage with and learn mathematics. Writing out problems and solutions helps reinforce concepts and improve retention.
- **Focus on key concepts:** College courses are often much more intensive than high school classes, and it is not feasible to master every detail. Concentrate on understanding the core ideas and principles, and don't get overwhelmed by the technical details.

- **Understand the "why":** In mathematics, understanding the underlying reasons and logic behind methods is more important than just knowing how to do computations. The "why" helps you grasp the broader implications and applications of the techniques you learn.
- **Gain practical experience:** Although this course emphasizes theoretical understanding and does not require programming, experimenting with statistical software such as R can be highly beneficial. Practical experience with data manipulation and analysis will enhance your comprehension and stimulate interest in the subject.
- **Engage with the material:** I will strive to make the course engaging and less boring. However, if this course is not your primary interest, focus on the aspects of the material that intrigue you. Try to have a general impression of the major concepts even though you do not remember any detail.
- **Exams are important, but more important is to enjoy the course.**

Recommended textbooks:

- DeGroot, M.H., and Schervish, M.J. (2011). *Probability and Statistics.*
- Hong, Y.M. (2017). *Probability Theory and Statistics for Economists.*

These textbooks are recommended references for this course, but you are not required to read them in their entirety. Focus on the lecture notes and exercises provided, and use these textbooks as supplementary resources as needed.

This content of this book is organized or follows. We start with probabilities based on counting, which should be familiar to high school graduates. Though rudimentary, they often yield surprising results, as rigorous calculations frequently challenge our intuitions about the likelihood of events. Special emphasis is placed on conditional probability, as conditional thinking is crucial both in academic studies as well as in daily life.

Next, we introduce the core concept of the random variable, which forms the foundation of all probability distributions and statistical theory. Random variables are essential tools that allow us to mathematically model uncertainty. We introduce two types of random variables: discrete and continuous. We begin

with discrete random variables because they do not require calculus, offering a smoother learning curve for beginners. Key concepts such as expectations, variance, and covariance are introduced alongside well-known discrete distributions such as the Binomial, Geometric, and Poisson distributions. This arrangement ensures that students can grasp these important concepts without being overwhelmed by calculus. We also demonstrate how these fundamental distributions can be applied to solve real-world problems.

Following this, we move on to continuous distributions. We will see that the formulas from discrete distributions extend naturally to continuous distributions with the aid of calculus—essentially replacing summation with integration. We cover some of the most important continuous distributions, such as the Normal, Exponential, and Gamma distributions, and explore the interconnections between them. We also extend the concepts of expectations, variance, and joint distributions to their continuous forms.

The book concludes with a discussion on sampling and statistical inference. Since we cannot observe entire distribution, it becomes necessary to infer distribution properties from finite samples. We introduce two of the most important theorems in probability and statistical theory—the Law of Large Numbers and the Central Limit Theorem. The breadth and generality of these theorems are remarkable. But their most significant contribution to statistical applications is they allow us to gauge how close our sample estimates are to the true parameter values. The final chapter also includes a brief discussion on estimator accuracy, confidence intervals, and hypothesis testing. These topics are introduced briefly, as they serve primarily to prepare students for more advanced courses, such as econometrics.

The chapters are organized logically, with each chapter building on the knowledge presented in the previous ones. Therefore, it is recommended to follow the sequence of chapters rather than reading them independently. However, advanced readers who are already familiar with the topics may feel free to skip between chapters as needed. This manuscript is written tersely, serving as a skeleton to complement lecture materials. It is not intended as a substitute for lectures or comprehensive textbooks. Students who wish to learn the course material solely by reading are encouraged to consult a formal textbook.

This manuscript is a preliminary version, and while efforts have been made to ensure accuracy, errors may still be present. Your feedback on any mistakes or

inaccuracies is greatly appreciated and will help improve the material.

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Chapter 1

Probability basics

Probability is the most important concept in modern science, especially as nobody has the slightest notion of what it means. ——Bertrand Russell

1.1 Definition of probability

1.1.1 High-school probability

We begin the discussion of probability from the most intuitive sense, where we forego the formal definition of "event" or "probability" for the moment. We will define these terms later on.

Definition 1.1. Naïve definition of probability:

$$P(A) = \frac{\#\text{ favorable outcomes}}{\#\text{ possible outcomes}}$$

assuming the outcomes are *finite* and *equally likely.*

Example 1.1. Flip a coin twice. Find the probability of landing two heads.

Solution: There are four possible outcomes: HH, HT, TH, TT, each with equal probability. Therefore, $P(\text{HH}) = \frac{1}{4}$.

Review of counting methods

- Multiplication rule (n_1 possible outcomes for the 1st experiment, n_2 possible outcomes for the 2nd experiment, ... then there are $n_1 n_2 \cdots n_r$ overall possible outcomes)
- Permutation (for ordered outcomes): $P_n^k = \frac{n!}{(n-k)!}$
- Combination (for unordered outcomes): $C_n^k = \frac{n!}{(n-k)!k!} := \binom{n}{k}$

	order matters	order doesn't matter
with replacement	n^k	C_{n+k-1}^k
non-replacement	P_n^k	C_n^k

Example 1.2. Find the probability of a "full house" in a five-card hand.

Solution:

$$P(\text{Full House}) = \frac{13C_4^3 \cdot 12C_4^2}{C_{52}^5} = 0.14\%.$$

Example 1.3 (Birthday problem)**.** Suppose there are k people. Find the probability that two of them have the same birthday.

Solution: Assuming there are 365 days in a year, ignoring leap years. If $k > 365$, the probability is 1. If $k \leq 365$,

$$P(\text{no match}) = \frac{365 \cdot 365 \cdots (365 - k + 1)}{365^k};$$

$$P(\text{match}) = \begin{cases} 50.7\% & k = 23 \\ 97\% & k = 50 \\ 99.999\% & k = 100 \end{cases}.$$

1.1.2 Axiomatic definition of probability

Definition 1.2. A **sample space** is the set of all possible outcomes. An **event** is the subset of the sample space.

Definition 1.3. A **probability space** consists of S and P, where S is a sample space, and P is a function which takes an event $A \subseteq S$ as input and returns $P(A) \in [0, 1]$ such that

- $P(\phi) = 0$,
- $P(S) = 1$,
- $P(\cup_{n=1}^{\infty} A_n) = \sum_{n=1}^{\infty} P(A_n)$ if $A_1, A_2, \ldots, A_n$ are disjoint.

Remark (Interpretations of probability). If we were to attach meaning to the word "probability", there are three approaches.

- **Classical approach**: It is based on the concept of *equally likely* outcomes. If there are n outcomes, then the probability of each outcome is $1/n$.
- **Frequency approach**: Repeat an experiment for n times and account the number of times that an event A occurs, say m. When n is sufficiently large, the relative frequency m/n may converge to a certain number, which we call the probability of A.
- **Subjective approach**: Probability is just the method that we use to make inferences when information is incomplete. It is our subjective belief about the likelihood of an event.

Note that this Definition 1.3 does not imply any particular interpretation of probability. In fact, any function P that satisfies the axioms are valid "probabilities". Thus, the theories of probability do not depend on any particular interpretation. It is purely axiomatic. From the three axioms, we can derive any property of probabilities. The interpretation also matters, but it is more of a philosophical debate.

Theorem 1.1. *Properties of probability:*

1. $P(A^C) = 1 - P(A)$
2. If $A \subseteq B$, then $P(A) \leq P(B)$.
3. $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

These properties can be easily justified using the **Venn diagram** (visualizing the sample space as an area in space). They can also be easily proved by the axioms. We will prove the second property and leave the rest as an exercise. The key is to break up the set into disjoint sets. If $A \subseteq B$, then $B = A \cup (B \cap A^c)$

where A and $B \cap A^c$ are disjoint. By the axioms in 1.3, we have $P(B) = P(A \cup (B \cap A^c)) = P(A) + P(B \cap A^c) \geq P(A)$.

The last property, regarding the union of events, can be generalized to three events:

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C);$$

and more generally,

$$P(A_1 \cup A_2 \cdots \cup A_n) = \sum_{j=1}^{n} P(A_j) - \sum_{i<j} P(A_i \cap A_j) + \sum_{i<j<k} P(A_i \cap A_j \cap A_k) - \cdots$$
$$(-1)^{n+1} P(A_1 \cap \cdots \cap A_n).$$

Example 1.4 (Matching problem)**.** Given n cards, labeled $1, 2, ..., n$. Let A_j be the event "j-th card matches"(the j-th card is numbered as j). Find the probability of at least one match, i.e. $P(A_1 \cup A_2 \cup \cdots \cup A_n) =?$

Solution: Since all position are equally likely, $P(A_j) = \frac{1}{n}$. The probability of there being two matches is: $P(A_1 \cap A_2) = \frac{(n-2)!}{n!} = \frac{1}{n(n-1)}$. Similarly, the probability of there being k matches is: $P(A_1 \cap \cdots \cap A_k) = \frac{(n-k)!}{n!} = \frac{1}{n(n-1)\cdots(n-k+1)}$. Using the property of the union of events,

$$\begin{aligned} P(A_1 \cup A_2 \cup \cdots \cup A_n) =& n \cdot \frac{1}{n} - \binom{n}{2} \frac{1}{n(n-1)} + \binom{n}{3} \frac{1}{n(n-1)(n-2)} - \cdots \\ =& 1 - \frac{1}{2!} + \frac{1}{3!} - \frac{1}{4!} + \cdots + (-1)^{n+1} \frac{1}{n!} \approx 1 - \frac{1}{e}. \end{aligned}$$

1.2 Conditional probability

Abraham Wald, the renowned statistician, was hired by the Statistical Research Group (SRG) at Columbia University to figure out how to minimize the damage to bomber aircraft. The data they had comprised aircraft returning from missions with bullet holes on their bodies. If asked which parts of the aircraft should be armored to enhance survivability, the obvious answer seemed to be to

armor the damaged parts. However, Wald suggested the exact opposite—to armor the parts that were not damaged. Why? Because the observed damage was conditioned on the aircraft returning. If an aircraft had been damaged on other parts, it likely would not have returned. Thinking conditionally completely changes the answer![1]

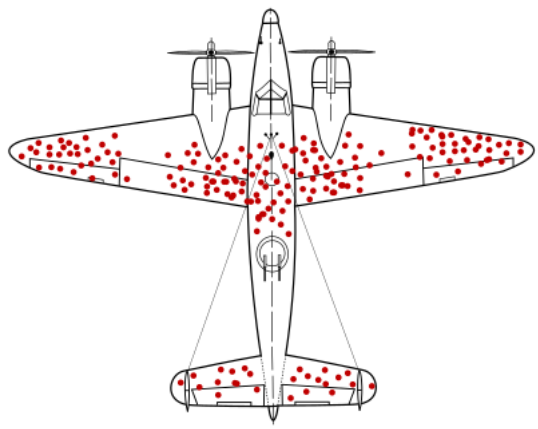


Figure 1.1: The damaged portions of returning planes show locations where they can sustain damage and still return home; those hit in other places presumedly do not survive (Image shows hypothetical data). See https://en.wikipedia.org/wiki/Survivorship_bias.

The probability of A **conditioned on** B is the updated probability of event A after we learn that event B has occurred. Since events contain information, the occurring of a certain event may change our believes on probabilities of other relevant events. The updated probability of event A after we learn that event B has occurred is the conditional probability of A given B.

Definition 1.4. If A and B are events with $P(B) > 0$, then the **conditional probability** of A given B is defined as

$$P(A|B) = \frac{P(A \cap B)}{P(B)}.$$

Exercise 1.1. Prove that conditional probabilities are probabilities. (Hint: using the three axioms.)

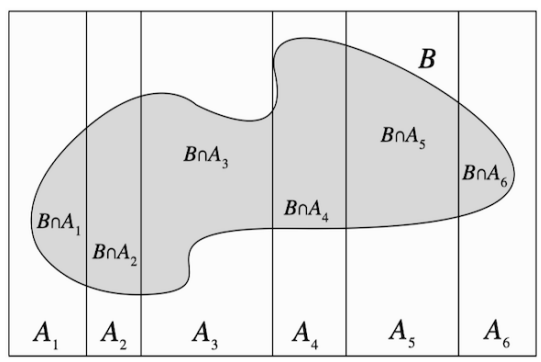
[1] See an interesting talk by Professor Joseph Blitzstein: "The Soul of Statistics". Available on http://www.youtube.com/watch?v=dzFf3r1yph8

Theorem 1.2. *Properties of conditional probability:*

- $P(A \cap B) = P(B)P(A|B) = P(A)P(B|A)$
- $P(A_1 \cap \cdots \cap A_n) = P(A_1)P(A_2|A_1)P(A_3|A_1, A_2) \cdots P(A_n|A_1 \ldots A_{n-1})$
- $P(A|B) = \dfrac{P(B|A)P(A)}{P(B)}$ *(Bayes' rule)*

Theorem 1.3 (Law of total probability)**.** *Let $A_1, ..., A_n$ be a partition of the sample space S (i.e., the A_i are disjoint events and their union is S), with $P(A_i) > 0$ for all i. Then*

$$P(B) = \sum_{i=1}^{n} P(B|A_i)P(A_i).$$



Example 1.5. Get a random 2-card hand from a standard deck. Find the probability of getting another ace conditioned on (a) having one ace, or (b) having the ace of spade.

Solution: The example shows the subtleness of conditional probabilities. The seemingly indifferent probabilities are in fact different:

$$P(\text{another ace} \mid \text{one ace}) = \frac{P(\text{both aces})}{P(\text{one ace})} = \frac{C_4^2/C_{52}^2}{1 - C_{48}^2/C_{52}^2} = \frac{1}{33};$$
$$P(\text{another ace} \mid \text{ace of spade}) = \frac{P(\text{ace of spade \& another ace})}{P(\text{ace of spade})} = \frac{C_3^1/C_{52}^2}{C_{51}^1/C_{52}^2} = \frac{1}{17}.$$

In the first case, the denominator is interpreted as "at least one ace"; whereas in the second case, it is "ace of space + another card".

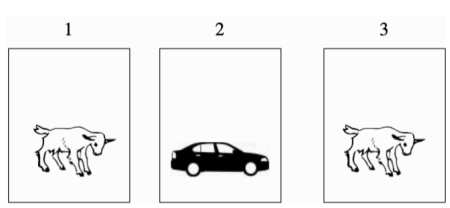
Example 1.6. The pandemic afflicted roughly 1/3 of the world population. The PCR test is 98% accurate. (this means if you have been infected, the test reports positive 98% of the time.) Find the probability of being infected when a test is positive.

Solution: Let D: actually infected, T: test positive. The test accuracy means: $P(T|D) = 98\%$. It also means $P(T|D^C) = 2\%$. We also know that $P(D) = 1/3$. We want to find $P(D|T)$. Apply the Bayes' rule:

$$\begin{aligned} P(D|T) =& \frac{P(T|D)P(D)}{P(T)} \\ =& \frac{P(T|D)P(D)}{P(T|D)P(D) + P(T|D^C)P(D^C)} \\ =& \frac{0.98 \times 1/3}{0.98 \times 1/3 + 0.02 \times 2/3} \approx 96\%. \end{aligned}$$

Note that how $P(T|D)$ is different from $P(D|T)$, though confusing the conditionality is quite common in daily life. The difference is even pronounced if the disease is rare. Suppose $P(D) = 10\%$. Then $P(D|T) = 84\%$. A large difference from the test accuracy rate 98%!

Example 1.7 (Monty Hall problem)**.** Suppose you are on Monty Hall's TV show. There are three doors. One of them has a car behind it. The other two doors have goats. Monty knows which one has the car. Monty now asks you to pick one door. You will win whatever is behind the door. After you pick one door. Monty opens another door that shows a goat. Monty then asks you if you want to switch. Is it optimal to switch?



We present two solutions to the problem. The first one is using the law of total probability —

Let S: succeed assuming switch; D_j: door j has the car, $j \in 1, 2, 3$. Without loss of generality, assume the initial pick is Door 1. Monty will always open the

door with a goat. By the law of total probability,

$$\begin{aligned}P(S) = & \underbrace{P(S|D_1)}_{\text{switch from initial pick}} P(D_1) + \underbrace{P(S|D_2)}_{\text{Monty opens door 3}} P(D_2) + \underbrace{P(S|D_3)}_{\text{Monty opens door 2}} P(D_3) \\ =& 0 + 1 \times \frac{1}{3} + 1 \times \frac{1}{3} = \frac{2}{3}.\end{aligned}$$

The problem can also be solved using the Bayes' rule —

Let D_j: door j has the car; M_j: Monty opens door j, $j \in 1, 2, 3$. Assume the initial pick is Door 1. If Monty opens door 3, the probability of winning the car assuming switching is

$$\begin{aligned}P(D_2|M_3) =& \frac{P(M_3|D_2)P(D_2)}{P(M_3)} \\ =& \frac{P(M_3|D_2)P(D_2)}{P(M_3|D_1)P(D_1) + P(M_3|D_2)P(D_2) + P(M_3|D_3)P(D_3)} \\ =& \frac{1 \times \frac{1}{3}}{\frac{1}{2} \times \frac{1}{3} + 1 \times \frac{1}{3} + 0} = \frac{2}{3}.\end{aligned}$$

Note that, if door 1 has the car, Monty will open door 2 and 3 with equal probability, thus $P(M_3|D_1) = \frac{1}{2}$. And Monty will never open the door with the car, therefore $P(M_3|D_3) = 0$. Similarly, if Monty opens door 2, we have $P(D_3|M_2) = \frac{2}{3}$. Therefore, the optimal choice is always to switch. Intuitively, because Monty knows which door has the car, the fact that he always opens the door without the car gives additional information regarding the choice of the door.

Example 1.8 (Simpson's paradox)**.** There are two doctors, Dr. Lee and Dr. Wong, performing two types of surgeries —heart surgery (hard) and band-aid removal (easy). Dr. Lee has higher overall surgery success rate. Is Dr. Lee necessarily a better doctor than Dr. Wong?

Solution: No. Consider the following example:

	Dr. Lee			Dr. Wong		
	Heart	Band-Aid	Total	Heart	Band-Aid	Total
Success	2	81	83	70	10	80
Failure	8	9	17	20	0	20
Success rate	20%	90%	83%	78%	100%	80%

The truth is Dr. Lee has overall higher success rate because he only does easy surgeries (band-aid removal). Dr. Wong does mostly hard surgeries and thus has a lower overall success rate. Yet, he is better at each single type of surgery. To formalize the argument, let S: successful surgery; D: treated by Dr. Lee, D^c: treated by Dr. Wong; E: heart surgery, E^c: band-aid removal. Dr. Wong is better at each type of surgery,

$$P(S|D,E) < P(S|D^c,E)$$
$$P(S|D,E^c) < P(S|D^c,E^c);$$

But, Dr. Lee has a higher overall successful rate,

$$P(S|D) > P(S|D^c).$$

This is because there is a "confounder" E:

$$P(S|D) = \underbrace{P(S|D,E)}_{<P(S|D^c,E)}\underbrace{P(E|D)}_{\text{weight}} + \underbrace{P(S|D,E^c)}_{<P(S|D^c,E^c)}\underbrace{P(E^c|D)}_{\text{weight}}.$$

A **confounder** is a variable that influences with both explanatory variable and the outcome variable, which therefore "confounds" the correlation between the two. In our example, the type of surgery (E) is associated with both the doctor and the outcome. Without the confounder being controlled, it is impossible to draw valid conclusions from the statistics.

In general terms, Simpson's paradox refers to the paradox in which a trend that appears across different groups of aggregate data is the reverse of the trend that appears when the aggregate data is broken up into its components. It is one of the most common sources of statistical misuse. Here is another example.[2]

Example 1.9 (UC Berkeley gender bias)**.** One of the best-known examples of Simpson's paradox comes from a study of gender bias among graduate school admissions to University of California, Berkeley. The admission figures for the fall of 1973 showed that men applying were more likely than women to be admitted, and the difference was so large that it was unlikely to be due to chance.

[2]See https://setosa.io/simpsons for a really good illustration of the Simpson's paradox.

	Male		Female	
	Applicants	Admitted	Applicants	Admitted
Total	8,442	44%	4,321	35%

However, when taking into account the information about departments being applied to, the conclusion turns to the opposite: in most departments, the admission rate for women is higher than men. The lower overall admission rate is caused by the fact that women tended to apply to more competitive departments with lower rates of admission, whereas men tended to apply to less competitive departments with higher rates of admission.

Department	Male		Female	
	Applicants	Admitted	Applicants	Admitted
A	825	62%	108	82%
B	560	63%	25	68%
C	325	37%	593	34%
D	417	33%	375	35%
E	191	28%	393	24%
F	373	6%	341	7%
Total	2691	45%	1835	30%

1.3 Independence

Definition 1.5. If event B's occurrence does not change the probability of A, then we say A and B are independent. That is to say A and B are **independent** if

$$P(A|B) = P(A) \text{ when } P(B) > 0.$$

Or more generally, A and B are **independent** if

$$P(A \cap B) = P(A)P(B).$$

(A definition including cases where A or B has zero probability.)

Theorem 1.4. *If events A and B are independent, then*

- *A and B^c are independent;*
- *A^c and B^c are independent.*

A and B are independent means they do not provide information to each other in the sense that conditional probability is not different from the unconditional probability. It is not an intuitive idea as it seems. It will become clearer when we discuss random variables in later chapters. Here we clarify some likely confusions.

Remark 1.1. Independence is not the same as disjointness.

A and B are disjoint means if A occurs, B cannot occur. But independence means A occurs has nothing to do with B.

Remark 1.2. Pairwise independence does not imply independence.

Definition 1.6. Events A, B, and C are said to be **(mutually) independent** if all of the following equations hold:

$$\begin{aligned} P(A \cap B) =& P(A)P(B), \\ P(A \cap C) =& P(A)P(C), \\ P(B \cap C) =& P(B)P(C), \\ P(A \cap B \cap C) =& P(A)P(B)P(C). \end{aligned}$$

If the first three conditions hold, we say that A, B, and C are **pairwise independent**. Pairwise independence does not imply independence. Convince yourself with the following example.

Example 1.10. Consider two fair, independent coin tosses, and let A be the event that the first is Heads, B the event that the second is Heads, and C the event that both tosses have the same result. Show that A, B, and C are pairwise independent but not independent.

Solution: For each event, $P(A) = \frac{1}{2}$, $P(B) = \frac{1}{2}$. Consider the two events together, there are four possible outcomes: HH, HT, TH, TT. $P(C) = P(HH)+$

$P(TT) = \frac{1}{2}$. Thus,

$$P(A \cap B) = P(HH) = \frac{1}{4} = P(A)P(B)$$
$$P(A \cap C) = P(HH) = \frac{1}{4} = P(A)P(C)$$
$$P(B \cap C) = P(HH) = \frac{1}{4} = P(B)P(C)$$

But A, B, C are not independent, because

$$P(A \cap B \cap C) = P(HH) = \frac{1}{4} \neq P(A)P(B)P(C).$$

Definition 1.7. For n events $A_1, A_2, \ldots, A_n$ to be **(mutually) independent**, we require any pair to satisfy $P(A_i \cap A_j) = P(A_i)P(A_j)$ (for $i \neq j$), any triplet to satisfy $P(A_i \cap A_j \cap A_k) = P(A_i)P(A_j)P(A_k)$ (for i, j, k distinct), and similarly for all quadruplets, quintuplets, and so on.

Definition 1.8. Events A and B are **conditional independent** given C if

$$P(A \cap B|C) = P(A|C)P(B|C).$$

Remark 1.3. Conditional independence does not apply independence.

Consider an example of playing chess games. Conditioned on the strength of your opponents, the outcome of each game is reasonably independent (ignoring the psychology and fatigues of the players). But the outcomes are not unconditionally independent, because stronger player has higher chances of winning each game.

Remark 1.4. Independence does not apply conditional independence.

Consider an example of fire alarm. Suppose there are two potential causes to trigger the fire alarm: (1) there is fire; (2) someone smoking. Assume the two events are independent. But they are not conditional independent if conditioning on the alarm beeping. Because if the alarm is on, but no one smokes, we definitely know there is fire. So there they are not conditional independent.

Chapter 2

Discrete random variables

2.1 Introduction to random variables

In the previous chapter, we have been working with *events*, which is a conceptualization of real world outcomes occurred with probabilities. In this chapter, we introduce a much more powerful conceptualization that deals with uncertain outcomes — random variables, which is the foundation of all probability and statistical studies.

In high school, all mathematical models come with certainty. For example, the falling time of any object from height h down to the earth is: $t = \sqrt{\frac{2h}{g}}$, where g is the gravity constant. The outcome is *deterministic*. The variables that enter into the equation either have unknown values or known certain values. Errors are possible only due to frictions or measurement errors.

But many real world processes come naturally with uncertainty. Think about the temperature tomorrow, or the stock market returns. We can only make predictions with probabilities. Yes, you may argue this uncertainly is due to incomplete information. If we have all the knowledge regarding the climate, we can predict exactly the temperature. But given the imperfection of the human knowledge, the only feasible option is to build this uncertainly into our mathematical models. Random variable is core concept and the Swiss knife that we use to deal with uncertainties mathematically.

Definition 2.1. Given an experiment with sample space S, a **random variable** is a function from the sample space S to the real numbers $\mathbb{R}$.

As an example, flipping a coin twice, let X be the number of heads. X could be $0, 1, 2$. X is a mapping from the sample space to real numbers. But it is unnatural to think of variables as functions. Here is an informal definition of random variable —

Definition (Informal)**.** A random variable is a variable whose value is uncertain, but comes with probabilities.

A random variable, say X, is never associated with a certain value, such as $X = 1$, or $X = 2$. It could be any of these values, but with different probabilities, e.g. $P(X = 1) = 0.2$, $P(X = 2) = 0.4$. Here are some examples of using random variables to describe uncertain outcomes.

Example 2.1 (Discrete random variable)**.** Given a 5-card hand, find the probability distribution of the number of aces. Let X be the number of aces. We have

$$P(X = k) = \begin{cases} \frac{C_4^k C_{48}^{5-k}}{C_{52}^5} & \text{if } k \in 0, 1, 2, 3, 4 \\ 0 & \text{otherwise} \end{cases}.$$

Each possible outcome is associated with a probability:

X	0	1	2	3	4
P	0.659	0.299	0.040	0.002	0.000

This is called a **distribution table**. A **distribution** is the mathematical function that gives the probabilities of occurrence of possible outcomes (formal definition in the next section).

In this example, because the values are discrete (there is no possible value between 0 and 1), we say X is a **discrete random variable**.

Example 2.2 (Continuous random variable)**.** Let X be the average temperature in September of Chengdu. We cannot predict X with certainty given the current technology. But if we do want an answer, we can think of X as a random variable with a continuous range of values associated with probabilities. History shows that $20 \leq X \leq 25$ is the mostly likely range.

In this example, X is a **continuous random variable**.

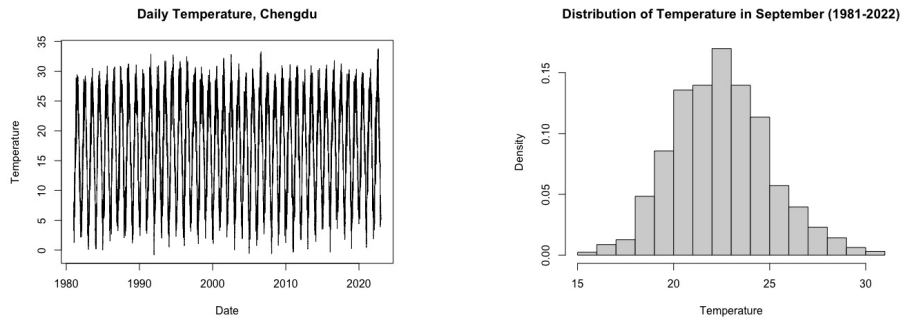


Figure 2.1: Temperature as a continuous random variable

Example 2.3 (Continuous random variable)**.** Suppose we want to predict daily stock market returns. Again, with the imperfections of our knowledge, a prediction with certainty is not possible. One way is to view the return as a random variable, and try to figure out the distribution of the random variable. The example shows the daily returns of the German DAX index. But other stock markets have similar patterns. The returns are mostly centered around zero, meaning that we cannot meaningfully predict the direction of the stock market by just looking at the history data.

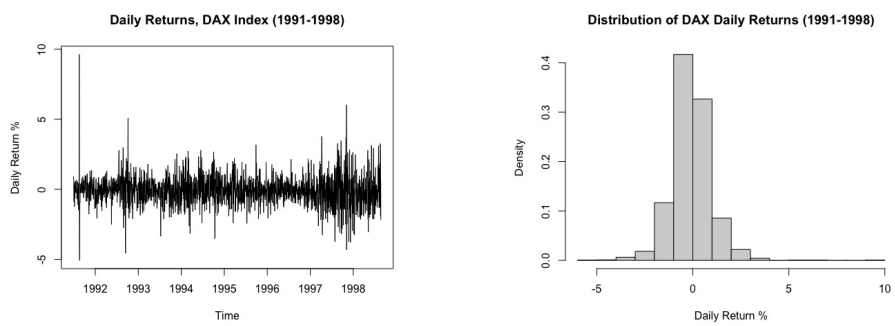


Figure 2.2: Stock market return as a continuous random variable

Remark (Sampling)*.* We notice that, in the above example, if we model the average temperature as a random variable X, we do not actually know the full distribution of X. History only gives a subset of the distribution. Is it possible that $X = 45$? The probability may not be exactly 0. It is just that it does not show up in our historical record.

When we conceptualize some real world events as random variables, the true distributions are mostly unknown. The data we observe is only a finite **sample** of that random variable. We can only approximate the true distribution with the observed frequencies and infer the properties of that random variable from the properties of the sample — this is the science of statistics.

2.2 Distributions and probability functions

Now we formally define the distribution of a random variable.

Definition 2.2. Let X be a random variable. The **distribution** of X is the collection of all probabilities of the form $P(X \in C)$ for all sets C of real numbers such that $X \in C$ is an event.

We already know there are two types of distributions: (1) **discrete distributions** and (2) **continuous distribution**.

2.2.1 Discrete distributions

Definition 2.3. We say that a random variable X has a **discrete distribution** if X can take only a finite number k of different values $x_1, \ldots, x_k$ or, at most, an infinite sequence of countable different values $x_1, x_2, \ldots$.

The finite or countably infinite set of values x such that $P(X = x) > 0$ is called the **support** of X.

Definition 2.4. If a random variable X has a discrete distribution, the **probability mass function** (PMF, sometimes also known as **probability function**, or **frequency function**) of X is defined as the function p such that

$$p(x) = P(X = x)$$

where $p(x) > 0$ for all possible values of x and $\sum_{\text{all } x} p(x) = 1$.

Examples of discrete distributions, including Bernoulli distribution and Binomial distribution, will be introduced in the following sections.

2.2.2 Continuous distributions

Definition 2.5. We say a random variable X has a **continuous distribution** if the possible values of X takes the form of a continuum.

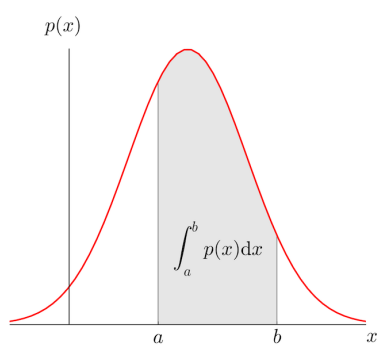
Definition 2.6. For a continuous random variable X, the **probability density function** (PDF) of X is a real-valued function f such that

$$P(a \leq X \leq b) = \int_a^b f(x)dx$$

where $f(x) \geq 0$ for all x and $\int_{-\infty}^{+\infty} f(x)dx = 1$.

Continuous random variables are usually measurements. Examples include height, weight, temperature, the amount of money and so on.

The probability of a continuous random variable is not defined at specific values. Instead, it is represented by **the area under a curve** (in advanced mathematics, this is known as an integral). The probability of observing any single value is equal to 0, since the number of values assumed by the random variable is infinite. For continuous distributions, $P(a < x < b) = P(a \leq x \leq b) = P(a \leq x < b) = P(a < x \leq b)$.



The simplest continuous distribution is the **Uniform distribution**. If X is distributed uniformly on interval $[a, b]$, it means X takes any value on the interval with equal chance. The PDF of X is given by

$$f(x) = \begin{cases} \frac{1}{b-a} & \text{if } a \leq x \leq b \\ 0 & \text{if } x < a \text{ or } x > b \end{cases}$$

We can verify that the area under the curve is 1.

More on continuous distributions will be discussed in the next chapter.

2.2.3 Cumulative distribution function

Unlike PMF or PDF, a cumulative distribution function can be defined for both discrete and continuous random variables.

Definition 2.7. The **cumulative distribution function** (CDF) of a random variable X is the function F given by $F(x) = P(X \leq x)$.

For discrete random variables, $F(x) = \sum_{k \leq x} p(k)$.

For continuous random variables, $F(x) = \int_{-\infty}^{x} f(t)dt$. We thus have $\frac{dF(x)}{dx} = f(x)$.

Example 2.4. In Example 2.1, the cumulative distribution function is given by

x	0	1	2	3	4
PMF	0.659	0.299	0.040	0.002	0.000
CDF	0.659	0.958	0.998	1.000	1.000

Theorem 2.1. Any CDF has the following properties:

- Increasing: if $x_1 \leq x_2$, then $F(x_1) \leq F(x_2)$.
- Right-continuous: for any a, $F(a) = \lim_{x \to a+} F(x)$.
- $F(x) \to 0$ as $x \to -\infty$; $F(x) \to 1$ as $x \to +\infty$.
- $P(X > x) = 1 - F(x)$
- $P(x_1 < x \leq x_2) = F(x_2) - F(x_1)$
- For each value x, $P(X < x) = F(x^-)$
- For every value x, $P(X = x) = F(x) - F(x^-)$

2.3 Bernoulli distribution

Definition 2.8. A random variable X is said to have the **Bernoulli distribution**, denoted as $X \sim \text{Bern}(p)$, if X has only two possible values, 0 and 1, and $P(X = 1) = p$, $P(X = 0) = 1 - p$.

The probability mass function p of a Bernoulli distribution is given by

$$p(x) = \begin{cases} p & \text{if } x = 1, \\ 1 - p & \text{if } x = 0. \end{cases}$$

This can also be expressed as

$$p(x) = p^x (1 - p)^{1-x} \text{ for } x \in 0, 1.$$

Example 2.5. Flip a coin once. Let X be the number of heads up. Then $X \sim \text{Bern}(p)$. If the coin is fair, we have $p = 0.5$.

2.4 Binomial distribution

Definition 2.9. Suppose $X_1, X_2, \ldots, X_n$ are independent and identical $\text{Bern}(p)$ distributions. Let X be the total number of successes of the n independent trials. That is, $X = X_1 + X_2 + \cdots + X_n$. Then X has the **Binomial distribution**, $X \sim \text{Bin}(n, p)$.

The probability mass function of X directly follows from the combination theory:

$$p(k) = P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}.$$

This is a valid PMF because, by the Binomial theorem, we have

$$\sum_{k=0}^{n} p(k) = \sum_{k=0}^{n} \binom{n}{k} p^k (1 - p)^{n-k} = (p + (1 - p))^n = 1.$$

The Binomial distribution is used to model the probability that a certain number of "successes" occur during a certain number of trials.

Example 2.6. Suppose it is known that 5% of adults who take a certain medication experience negative side effects. Find the probability that a certain number of patients in a random sample of 100 will experience negative side effects.

Solution: Let X be the number patients that experience negative side effects, then $X \sim \text{Bin}(100, 0.05)$. Computing the probabilities:

$$P(X > k) = \begin{cases} 0.3840 & k = 5 \\ 0.0115 & k = 10 \\ 0.0004 & k = 15 \end{cases}$$

So it is very unlikely that more than 10 patients would have negative side effects.

Example 2.7. Suppose you are shopping a product online. There are three sellers with the following ratings:

- Seller 1: 100% positive out of 10 reviews
- Seller 2: 96% positive out of 50 reviews
- Seller 3: 93% positive out of 200 reviews

Which seller is likely to give the best service?

Solution: The problem is intriguing because it is obvious that higher ratings do not necessarily means higher satisfaction. We have to weight in the number of reviews. The more reviews, the more trustworthy the ratings are. Let $X_j^{(i)}$ be a random variable that means consumer j is satisfied with seller i, where $i \in 1, 2, 3$. Assume $X_j^{(i)}$ follows a Bernoulli distribution:

$$X_j^{(i)} = \begin{cases} 1 & \text{satisfaction with probability } \theta_i \\ 0 & \text{otherwise} \end{cases}$$

where θ_i is an unknown parameter of seller i that captures their "genuine" satisfaction rate. We assume the consumers independently write their ratings. The overall positive rate of seller i is $S_i = \frac{1}{n_i} \sum_j X_j^{(i)}$ where n_i is the total number of reviews. We want to infer the value of θ_i from their observed positive rate S_i. From now on we drop the seller index i to simply the notation since it is symmetric for all sellers.

Because we have no prior knowledge about θ. We assume that θ takes any value from $[0, 1]$ equally likely, i.e. $\theta \sim \text{Unif}(0, 1)$. Assuming each X_j is independent and identical, then

$$S = X_1 + X_2 + \cdots + X_{n_i}$$

follows the Binomial distribution with PMF:

$$p(k|\theta) = \binom{n}{k} \theta^k (1-\theta)^{n-k}$$

Our goal is to find: $p(\theta|k)$. Recall that the Bayes' rule allows us to invert the conditional probability:

$$p(\theta|k) = \frac{p(k|\theta)p(\theta)}{p(k)} = \frac{p(k|\theta)p(\theta)}{\int_{-\infty}^{\infty} p(k|\theta)p(\theta)d\theta}$$

Since $\theta \sim \text{Unif}(0, 1)$, we have

$$p(\theta) = \begin{cases} 1 & \text{if } \theta \in [0, 1] \\ 0 & \text{otherwise} \end{cases}$$

We now focus on $\theta \in [0, 1]$, since the probability is 0 otherwise. Substitute in the PMF of the Binomial distribution,

$$p(\theta|k) = \frac{\binom{n}{k}\theta^k (1-\theta)^{n-k}}{\int_0^1 \binom{n}{k}\theta^k (1-\theta)^{n-k} d\theta}$$

The hard part is to evaluate the integral. We state without proof (you need to trust me on this at the moment):

$$\int_0^1 \theta^k (1-\theta)^{n-k} = \frac{k!(n-k)!}{(n+1)!}$$

Therefore,

$$p(\theta|k) = \frac{(n+1)!}{k!(n-k)!} \theta^k (1-\theta)^{n-k}$$

Now suppose you are the next customer. The probability that you would be satisfied is

$$\begin{aligned}
P(X_{n+1} = 1|S = k) &= \int_0^1 P(x_{n+1} = 1|\theta)p(\theta|k)d\theta \\
&= \int_0^1 \theta \times \frac{(n+1)!}{k!(n-k)!}\theta^k(1-\theta)^{n-k}d\theta \\
&= \frac{(n+1)!}{k!(n-k)!}\int_0^1 \theta^{k+1}(1-\theta)^{(n+1)-(k+1)}d\theta \\
&= \frac{(n+1)!}{k!(n-k)!} \times \frac{(k+1)!(n-k)!}{(n+2)!} \\
&= \frac{k+1}{n+2}.
\end{aligned}$$

Now we substitute the ratings for the three sellers:

- Seller 1: $n = 10, k = 10$
- Seller 2: $n = 50, k = 48$
- Seller 3: $n = 200, k = 186$

The probabilities that you would be satisfied with each seller are: 92%, 94%, 93%. The result is known as the **Laplace's rule of succession**. The rule of thumb is, pretending we have too more reviews: one is positive, the other is negative. Compute the satisfaction rate as $\frac{k+1}{n+2}$.

2.5 Expectation

Definition 2.10. Let X be a discrete random variable. The **expectation** of X, denoted by $E(X)$, is defined as:

$$E(X) = \sum_{\text{all } x} xP(X = x).$$

The expectation of X is also referred to as the **mean** of X or the **expected value** of X.

Example 2.8. The expectation of a Bernoulli random variable $X \sim \text{Bern}(p)$:

$$E(X) = 1 \times P(X = 1) + 0 \times P(X = 0) = p.$$

Example 2.9. The expectation of a Binomial random variable $X \sim \text{Bin}(n, p)$:

$$\begin{aligned}
E(X) =& \sum_{k=0}^{n} kp(k) \\
=& \sum_{k=0}^{n} k \cdot \binom{n}{k} p^k q^{n-k} \\
=& \sum_{k=1}^{n} n \cdot \binom{n-1}{k-1} p^k q^{n-k} \\
=& np \sum_{k=1}^{n} \binom{n-1}{k-1} p^{k-1} q^{n-k} \\
=& np \underbrace{\sum_{j=0}^{n-1} \binom{n-1}{j} p^j q^{n-1-j}}_{\text{another Binomial PMF}} \\
=& np.
\end{aligned}$$

Example 2.10. Life expectancy is the average number of years a person is expected to live. It is a crucial indicator of the quality of living and one of the three components of the Human Development Index (HDI) (the other two components are education and per capita GDP). Here is a toy example to compute life expectancy with hypothetical data.[1]

Age Group	No. of People	No. of Deaths	Mortality Rates	Surviving to Age	Number Surviving	Living to Exact Age
				0	1000	5
0-19	200	1	0.5%	20	995	3
20-39	300	1	0.3%	40	992	12
40-59	250	3	1.2%	60	980	78
60-79	150	12	8.0%	80	901	901
80-99	100	100	100%	100	0	0
Total	1000					

Table 2.1: Hypothetical mortality rates and life table

Note that life expectancy is **not** the average age of the population. Given the

[1] This is an overly simplified example that only serves to clarify the definition of expectation. See this tutorial from MEASURE Evaluation for the actual computation of life expectancy.

hypothetical data in the above table, if we take the midpoints of each age group and compute the weighted average,

$$\overline{\text{Age}} = (10 \times 200 + 30 \times 300 + 50 \times 250 + 70 \times 150 + 90 \times 100)/1000 = 43.$$

But this is not the number of years a person is expected to live. The computation does not even consider the chances of mortality at each age. Let X be a random variable that represents the number of years a person lives. Life expectancy should be computed as $E(X) = \sum_t tP(X = t)$.

Based on the hypothetical data, we compute the age-specific mortality rates. Suppose we have 1000 new-born babies. The chances that they would survive to age 20 is given by $1000 \times (1 - 0.5\%) = 995$. The changes that these people would survive to age 40 is $995 \times (1 - 0.3\%) \approx 992$. Similarly, we compute the rest of the surviving numbers.

To simplify the calculation, we make a bold assumption here that there is no survival between the cutoffs. For example, if a person does not survive to 40, he will die at the age 20. There is not age such as 25 or 30 in between. The number of people who die at age 20 would be those who survive the 0-19 period but does not survive the 20-39 period, which is $995 - 992 = 3$, or 0.3% of the initial population. The life expectancy is thus computed as

$$E(X) = 20 \times 0.3\% + 40 \times 1.2\% + 60 \times 7.8\% + 80 \times 90.1\% \approx 77.$$

Theorem 2.2 (Linearity)**.** *Expectations have the following properties regardless of the dependencies of the random variables:*

- $E(X + Y) = E(X) + E(Y)$
- If $Y = aX + b$, then $E(Y) = aE(X) + b$.
- $E(X_1 + \cdots + X_n) = E(X_1) + \cdots + E(X_n)$
- $E(a_1X_1 + \cdots + a_nX_n + b) = a_1E(X_1) + \cdots + a_nE(X_n) + b$

Example 2.11. Redo the expectation of $X \sim \text{Bin}(n, p)$ with properties of expectation:

$$E(X) = E(X_1 + \cdots + X_n) = nE(X_i) = np$$

where $X_i \sim \text{Bern}(p)$.

Example 2.12. Suppose that there are n people sitting in a classroom with exactly n seats. At some point, everyone got up, ran around the room, and sat back down randomly (i.e., all seating arrangements are equally likely). What is the expected value of the number of people sitting in their original seat?

Solution: Number the people from 1 to n. Let X_i be the Bernoulli random variable with value 1 if person i returns to their original seat and value 0 otherwise. Since person i is equally likely to sit back down in any of the n seats, the probability that person i returns to their original seat is $1/n$. Therefore $E[X_i] = 1/n$. Now, let X be the number of people sitting in their original seat following the rearrangement. Then $X = X_1 + X_2 + \cdots + X_n$. By linearity of expected values, we have $E[X] = \sum E[X_i] = \sum 1/n = 1$.

Theorem 2.3. If X and Y are *independent*, we have

$$E(XY) = E(X)E(Y).$$

In general, if $X_1, \ldots, X_n$ are *independent*, we have

$$E(X_1 X_2 \cdots X_n) = E(X_1)E(X_2)\cdots E(X_n).$$

<u>Note that the multiplication rule will not hold without independence</u>.

Theorem 2.4 (Law of the unconscious statistician (LOTUS))**.** *Let X be a random variable, and g be a real-valued function of a real variable. If X has a discrete distribution, then*

$$E[g(X)] = \sum_{all\ x} g(x)P(X = x).$$

Example 2.13. Compute $E(X)$ and $E(X^2)$ given the following distribution.

X	1	2	3	4
$P(X)$	0.2	0.3	0.3	0.2

Solution: According to the distribution table, we compute the expectation as

$$E(X) = 1 \times 0.2 + 2 \times 0.3 + 3 \times 0.3 + 4 \times 0.2 = 2.5$$

The distribution table for $Y = X^2$ is given by

Y	1	4	9	16
$P(Y)$	0.2	0.3	0.3	0.2

Therefore,

$$E(Y) = 1 \times 0.2 + 4 \times 0.3 + 9 \times 0.3 + 16 \times 0.2 = 7.3$$

Note that $E(X^2) \neq [E(X)]^2$.

Remark 2.1. In general, $E[g(X)] \neq g(E(X))$. Linearity implies that if g is a linear function of X, then $E[g(X)] = g(E(X))$. For a nonlinear function g, the relationship between $E[g(X)]$ and $g(E(X))$ is determined case by case. We will get back to this point when we learn Jensen's inequality.

Example 2.14 (St. Petersburg Paradox)**.** Flip a fair coin over and over again until the head lands the first time. You will win 2^k dollars if the head lands in the k-th trial (including the successful trial). What is the expected payoff of this game?

Solution: Let $X = 2^k$. We want to find $E(X)$. The probability of the first head showing up in the k-th trial is $\frac{1}{2^k}$. Therefore,

$$E(X) = \sum_{k=1}^{\infty} 2^k \cdot \frac{1}{2^k} = \sum_{k=1}^{\infty} 1 = \infty$$

The expected payoff is infinitely high! This is against most people's intuition. This is because we intuitively think that $E(X) = E(2^k) = 2^{E(k)}$, which is a finite number.

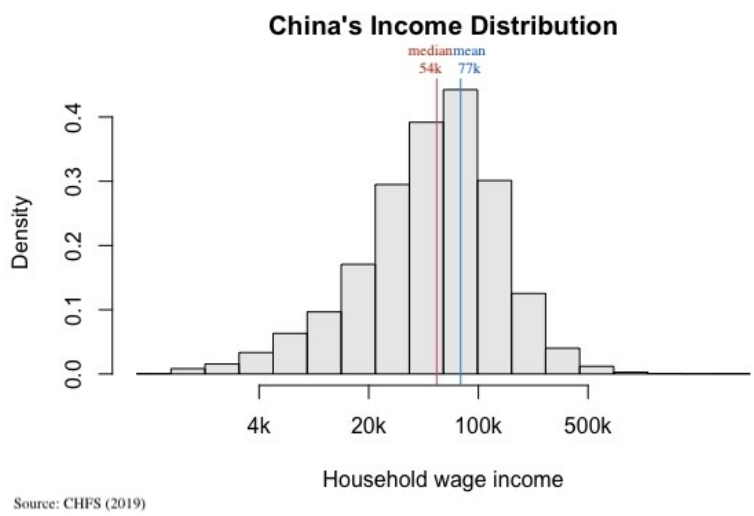
2.6 Mean and Median

Definition 2.11. Let X be a random variable. Every number m with the following property is called a **median** of X:

$$P(X \leq m) \geq \frac{1}{2}, \text{ and } P(X \geq m) \geq \frac{1}{2}.$$

Every distribution must have at least one median, but there may be multiple medians. (think about a distribution with even number of outcomes)

The main reason why the median is sometimes preferred over the mean is that the median is more robust to extreme values. Think about an income distribution. Higher incomes are rare. But their absolute values are high. Thus, the mean income tends be higher than what the mass of the population would earn. But the median is more robust to extreme values and is closer to the earnings of an "average" person. For example, the mean of China's household income is ¥77,000 annually; the median is only ¥54,000.



2.7 Variance and Covariance

Definition 2.12. Let X be a random variable with finite mean $\mu = E(X)$. The **variance** of X, denoted by $Var(X)$, is defined as

$$Var(X) = E\left[(X-\mu)^2\right].$$

The **standard deviation** of X is defined as $\sigma_X = \sqrt{Var(X)}$.

The variance (as well as the standard deviation) of a distribution provides a measure of the spread or dispersion of the distribution around its mean.

Theorem 2.5. *Variance has the following properties:*

- $Var(X) = E(X^2) - [E(X)]^2$
- $Var(X + c) = Var(X)$
- $Var(cX) = c^2 Var(X)$
- *In general,* $Var(X + Y) \neq Var(X) + Var(Y)$. *See the covariance part.*

Example 2.15. Find the variance for $X \sim \text{Bern}(p)$.

$$Var(X) = E(X^2) - E^2(X) = p - p^2 = p(1 - p).$$

Example 2.16. Find the variance for $X \sim \text{Bin}(n, p)$. $X = X_1 + \cdots + X_n$ where X_i are $i.i.d$ Bernoulli distributions.

$$Var(X) \overset{iid}{=} \sum_{i=1}^{n} Var(X_i) = np(1 - p).$$

Definition 2.13. Let X and Y be random variables with finite means $E(X) = \mu_X$ and $E(Y) = \mu_Y$. The **covariance** of X and Y, denoted by $Cov(X, Y)$, is defined as

$$Cov(X, Y) = E[(X - \mu_X)(Y - \mu_Y)].$$

The **correlation** of X and Y, the standardized covariance, is defined as

$$\rho_{XY} = \frac{Cov(X, Y)}{\sigma_X \sigma_Y}.$$

A correlation is always between $[-1, 1]$. It is said that X and Y are **positively correlated** if $\rho_{XY} > 0$, that X and Y are **negatively correlated** if $\rho_{XY} < 0$, and that X and Y are **uncorrelated** if $\rho_{XY} = 0$.

Covariances and correlations provide measures of the extend to which two random variables are linearly related. If we plot the values of X and Y in the xy-plane, if the points form a straight line, that would signal a strong positive (if positive slope) or negative (if negative slope) correlation. It is possible that the correlation is 0 if X and Y are dependent but the relationship is nonlinear (Figure 2.3).

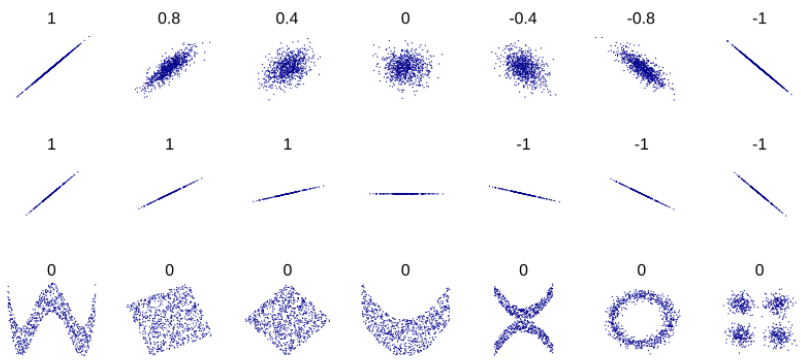


Figure 2.3: Types of dependencies and correlations

Theorem 2.6. *Properties of variances and covariances:*

- $Cov(X, Y) = E(XY) - E(X)E(Y)$
- $Cov(X, X) = Var(X)$
- $Cov(X, Y) = Cov(Y, X)$
- $Cov(cX, Y) = Cov(X, cY) = c\,[Cov(X, Y)]$
- $Cov(X + Y, Z) = Cov(X, Z) + Cov(Y, Z)$
- $Var(X + Y) = Var(X) + Var(Y) + 2Cov(X, Y)$
- $Var(aX + bY + c) = a^2Var(X) + b^2Var(Y) + 2abCov(X, Y)$
- $Var\left(\sum_{i=1}^{n} X_i\right) = \sum_{i=1}^{n} Var(X_i) + 2\sum_{i=1}^{n}\sum_{i<j} Cov(X_i, X_j)$
- $Var\left(\sum_{i=1}^{n} X_i\right) = \sum_{i=1}^{n} Var(X_i)$ *if* $X_1, \ldots, X_n$ *are uncorrelated.*

Theorem 2.7. *If* X, Y *are independent, they are uncorrelated. But the converse is false.*

Proof. $Cov(X, Y) = E(XY) - E(X)E(Y)$. Independence implies $E(XY) = E(X)E(Y)$. Thus, $Cov(X, Y) = 0$. But $Cov(X, Y) = 0$ does not necessarily imply independence. □

2.8 Moments*

Definition 2.14. Let X be a random variable and k a positive integer. The expectation $E(X^k)$ is called the k-th **moment** of X. The expectation $E[(X - E(X))^k]$ is called the k-th **central moment** of X (or the k-the moment of X about the mean).

In accordance with this terminology, $E(X)$ is the first moment of X, $Var(X)$ is the second central moment of X. It is natural to ask if there are higher order moments. The answer is yes.

Definition 2.15. Let X be a random variable with mean μ, standard deviation σ, and finite third moment. The **skewness** of X is defined as

$$\text{Skew}(X) = E\left[\left(\frac{X-\mu}{\sigma}\right)^3\right].$$

Definition 2.16. The **Kurtosis** of X is defined as

$$\text{Kurt}(X) = \left[\left(\frac{X-\mu}{\sigma}\right)^4\right].$$

Skewness is the measure of the lopsidedness of the distribution; any symmetric distribution will have a third central moment, if defined, of zero. A distribution that is skewed to the left (the tail of the distribution is longer on the left) will have a negative skewness. A distribution that is skewed to the right (the tail of the distribution is longer on the right), will have a positive skewness.

Kurtosis is a measure of the heaviness of the tail of the distribution. If a distribution has heavy tails, the kurtosis will be high; conversely, light-tailed distributions have low kurtosis.

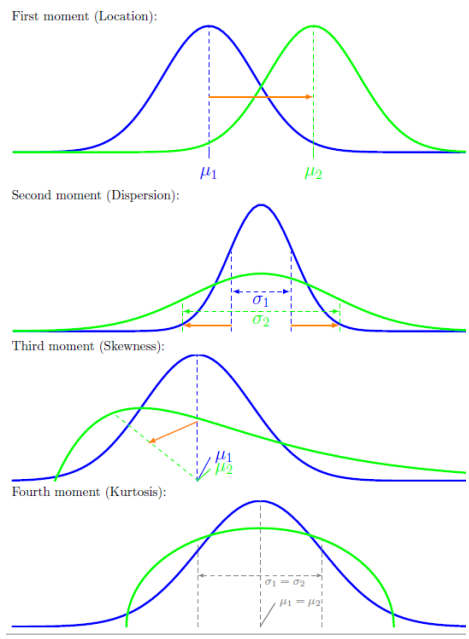


Figure 2.4: Moments and the shape of a distribution

We see that moments give information about the shape of a distribution. Different orders of moments captures different aspects of the distribution. In fact, if we know all the moments (moments of infinitely high order), we can exactly pin down the distribution.

Theorem 2.8. *For a distribution of mass or probability on a bounded interval, the collection of all the moments (of all orders, from* 0 *to* ∞*) uniquely determines the distribution.*

So there are two ways of fully characterize a distribution:

1. Listing all the possible values along with their associated probabilities;
2. Giving all the moments of the distribution.

It is somewhat like the analogous Taylor theorem in the probability theory. We can represent any distribution by a sequence of higher order "polynomials": $E(X), E(X^2), E(X^3), \ldots$

Definition 2.17. Let X be a random variable. For each real number t, define the **moment generating function** (MGF) as

$$M_X(t) = E\left(e^{tX}\right).$$

To see why it is "generating" moments, take the Taylor expansion of the exponential function:

$$e^{tX} = 1 + tX + \frac{t^2X^2}{2!} + \frac{t^3X^3}{3!} + \cdots$$

Hence,

$$M_X(t) = E\left(e^{tX}\right) = 1 + E(X)t + E(X^2)\frac{t^2}{2!} + \cdots$$

Theorem 2.9. *Let $M_X(t)$ be the MGF of X. Then the n-th moment of X is given by $E(X^n) = M_X^{(n)}(0)$, where $M_X^{(n)}$ denotes the n-th derivative of the MGF.*

2.9 Geometric and Negative Binomial

Definition 2.18. Consider a sequence of independent Bernoulli trials, each with the same success probability p. Let X be the number of failures before the first successful trial. Then X has a **Geometric distribution**, $X \sim \text{Geom}(p)$.

Let's derive the PMF for the Geometric distribution. By definition,

$$P(X = k) = q^k p$$

where $q = 1 - p$. This is a valid PMF because

$$\sum_{k=0}^{\infty} q^k p = p \sum_{k=0}^{\infty} q^k = \frac{p}{1-q} = 1.$$

The expectation of X is given by

$$E(X) = \sum_{k=0}^{\infty} k \cdot q^k p = p \sum_{k=0}^{\infty} k q^k = p\frac{q}{p^2} = \frac{1-p}{p}.$$

To see why this holds, taking derivative with respect to k on both sides of $\sum_{k=0}^{\infty} q^k = \frac{1}{1-q}$ yields

$$\sum_{k=1}^{\infty} kq^{k-1} = \frac{1}{(1-q)^2};$$

Then multiply both sides by q:

$$\sum_{k=1}^{\infty} kq^{k} = \frac{q}{(1-q)^2}.$$

A generalization of the Geometric distribution is the Negative Binomial distribution.

Definition 2.19. In a sequence of independent Bernoulli trials with success probability p, if X is the number of failures before the r-th success, then X is said to have a **Negative Binomial distribution**, denoted $X \sim \mathrm{NBin}(r, p)$.

The PMF for Negative Binomial distribution, by definition, is given by

$$P(X = k) = \binom{k+r-1}{r-1} q^k p^r.$$

To compute the expectation, let $X = X_1 + \cdots + X_r$ where X_i is the number of failures between the $(i-1)$-th success and the i-th success, $1 \le i \le r$. Then $X_i \sim \mathrm{Geom}(p)$. By linearity of expectations,

$$E(X) = E(X_1) + \cdots + E(X_r) = r\frac{1-p}{p}.$$

Example 2.17 (Toy collector)**.** There are n types of toys. Assume each time you buy a toy, it is equally likely to be any of the n types. What is the expected number of toys you need to buy until you have a complete set?

Solution: Define the following random variables:

$$\begin{aligned}
T =& T_1 + T_2 + \cdots + T_n \\
T_1 =& \text{number of toys until 1st new type} \\
T_2 =& \text{additional number of toys until 2nd new type} \\
T_3 =& \text{additional number of toys until 3rd new type} \\
&\vdots
\end{aligned}$$

We know, $T_1 = 1$, $T_2 - 1 \sim \text{Geom}\left(\frac{n-1}{n}\right)$,..., $T_j - 1 \sim \text{Geom}\left(\frac{n-(j-1)}{n}\right)$. Thus,

$$\begin{aligned}
E(T) =& E(T_1) + E(T_2) + \cdots + E(T_n) \\
=& 1 + \frac{n}{n-1} + \frac{n}{n-2} + \cdots + \frac{1}{n} \\
=& n(1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n}) \\
\to& n(\log n + 0.577).
\end{aligned}$$

2.10 Poisson distribution

Now we introduce arguably the most popular discrete distribution—Poisson distribution. Poisson distribution is used to model independent events occurring at a constant mean rate. It is like the Binomial distribution in the sense that they both model the number of occurrence of events, but it is parametrized on the "rate" of the event (how many times an event occurs in a unit of time on average) rather than the total number of events and the probability of each event. It is therefore more practical in real-world modeling since we mostly observe the rate rather than the totality. We introduce the Poisson distribution by showing that it is a limit case of the Binomial distribution.

Suppose a business has n customers. Each day there are m customers arriving for service. The probability of each customer arrives is therefore $p = \frac{m}{n}$. Let X be the number of customers arrive per day, we want to compute the probability distribution of X. It is tempting to model X as a Binomial distribution,

$$X \sim \text{Bin}(n, p)$$

However, Binomial distribution assumes each customer is independent Bernoulli, meaning each customer either arrives ($X_i = 1$) or not ($X_i = 0$) in a given day. This is an unreasonable assumption, since each customer can arrive multiple times in a day.

One way to fix the multiple-time problem is to make the time unit smaller, say, an hour. Then the probability for each customer to arrive per hour is $p' = \frac{p}{24}$. Let X' be he number of customers arrive per hour. If each customer either arrives or not per hour, then

$$X' \sim \text{Bin}(n, p')$$

It is better than the first model, but still not ideal. Since X_i' are not fully Bernoulli either. In fact, it is better to model customer arrivals as continuous. Let's divide the time unit to infinitesimal, thus $p \to 0$, while holding the "rate" $m = np$ (average arrivals per day) constant. This entails $n \to \infty$. It makes no sense to parametrize n, p anymore. Instead, we parametrize the rate $\lambda = np$. Take limit of the Binomial distribution:

$$\begin{aligned}
P(X = k) &= \lim_{n\to\infty} \binom{n}{k} p^k (1-p)^{n-k} \\
&= \lim_{n\to\infty} \binom{n}{k} \left(\frac{\lambda}{n}\right)^k \left(1 - \frac{\lambda}{n}\right)^{n-k} \\
&= \lim_{n\to\infty} \binom{n}{k} \left(\frac{\lambda}{n}\right)^k \underbrace{\left(1 - \frac{\lambda}{n}\right)^n}_{\to e^{-\lambda}} \underbrace{\left(1 - \frac{\lambda}{n}\right)^{-k}}_{\to 1} \\
&= \lim_{n\to\infty} \underbrace{\frac{n!}{(n-k)!k!}}_{\to 1} \frac{\lambda^k}{n^k} e^{-\lambda} \\
&= \frac{\lambda^k}{n^k} e^{-\lambda}.
\end{aligned}$$

This is the PMF of the Poisson distribution.

Definition 2.20. A random variable X has the Poisson distribution with parameter λ if the PMF of X is

$$P(X = k) = \frac{e^{-\lambda}\lambda^k}{k!}, \quad k = 0, 1, 2, \ldots$$

We denote this as $X \sim \text{Pois}(\lambda)$.

We can easily verify this is a valid PMF because $\sum_{k=0}^{\infty} \frac{\lambda^k}{k!} = e^{\lambda}$.

Poisson distribution is used to model the probability of a given number of events occurring in a fixed interval of time if these events occur with a known constant mean rate (#events / time unit).

The expectation of the Poisson distribution is

$$\begin{aligned}E(X) =& \sum_{k=0}^{\infty} k \cdot \frac{e^{-\lambda}\lambda^k}{k!} \\ =& e^{-\lambda} \sum_{k=1}^{\infty} \frac{\lambda^k}{(k-1)!} \\ =& \lambda e^{-\lambda} \sum_{k=1}^{\infty} \frac{\lambda^{k-1}}{(k-1)!} \\ =& \lambda e^{-\lambda} e^{\lambda} = \lambda.\end{aligned}$$

To get the variance, we first compute $E(X^2)$. By LOTUS,

$$E(X^2) = \sum_{k=0}^{\infty} k^2 \cdot \frac{e^{-\lambda}\lambda^k}{k!} = e^{-\lambda} \sum_{k=1}^{\infty} k^2 \frac{\lambda^k}{k!}$$

Differentiate $\sum_{k=0}^{\infty} \frac{\lambda^k}{k!} = e^{\lambda}$ on both sides with respect to λ and multiply (replenish) again by λ:

$$\sum_{k-1}^{\infty} k \frac{\lambda^k}{k!} = \lambda e^{\lambda}$$

Repeat:

$$\sum_{k-1}^{\infty} k^2 \frac{\lambda^k}{k!} = \lambda(e^{\lambda} + \lambda e^{\lambda})$$

Therefore, we have

$$E(X^2) = e^{-\lambda}(\lambda + \lambda^2)e^{\lambda} = \lambda + \lambda^2$$

Finally,

$$Var(X) = E(X^2) - (E(X))^2 = \lambda + \lambda^2 - \lambda^2 = \lambda.$$

Example 2.18. What is the probability of an earthquakes in a year in Sichuan?

Historical records[2] show that, from 26 BCE to 2021 CE, there were 309 earthquakes with magnitude of 5.0 or greater. Let X be the number of earthquakes

[2]See this article from the Sichuan Earthquake Administration.

with magnitude 5.0 or greater. The annual rate λ of earthquakes is therefore $\frac{309}{2048} = 0.15$. Assume earthquakes are independent events (not always the case). Then $X \sim \text{Pois}(0.15)$. By the distribution of the Poisson distribution,

$$P(X = k) = \begin{cases} 0.86 & k = 0 \\ 0.13 & k = 1 \\ 0.01 & k = 2 \end{cases}.$$

2.11 Joint, conditional and marginal distributions

We now consider the distribution of multi-variables. The simplest case is bivariate distributions.

Definition 2.21. Let X and Y be random variables. Consider the ordered pair (X, Y). If there are only finitely or at most countably many different possible values (x, y), we say that X and Y have a **discrete joint distribution**.

Definition 2.22. The **joint PMF** of X and Y is defined as the function p such that for every point (x, y),

$$p(x, y) = P(X = x, Y = y)$$

where $\sum_x \sum_y p(x, y) = 1$. The comma means the two conditions have to be satisfied at the same time.

Example 2.19. Let X be an indicator of an individual being a current smoker. Let Y be the indicator of his developing lung cancer at some point in his life. The joint PMF of X and Y is as specified in the table below.

	$Y = 1$	$Y = 0$	**Total**
$X = 1$	0.05	0.20	0.25
$X = 0$	0.03	0.72	0.75
Total	0.08	0.92	1

Definition 2.23. The **joint CDF** of two random variables X and Y is defined as the function F such that for all values of x and y,

$$F(x, y) = P(X \leq x, Y \leq y).$$

Definition 2.24. For discrete random variables X and Y, the **conditional PMF** of Y given $X = x$ is

$$p_{Y|X}(y|x) = \frac{P(X = x, Y = y)}{P(X = x)}$$

This is viewed as a function of y for fixed x.

Definition 2.25. For discrete random variables X and Y, the **marginal PMF** of X is

$$p_X(x) = \sum_{\text{all } y} P(X = x, Y = y)$$

That is, we *marginalize out* Y leaving only X.

Example 2.20. In the previous example, the conditional PMF of having lung cancer conditioned on being a smoker is

$$P(Y = 1|X = 1) = \frac{P(X = x, Y = y)}{P(X = x)} = \frac{0.05}{0.25} = \frac{1}{5}.$$

The marginal PMF for having lung cancer is

$$\begin{aligned} P(Y = 1) =& P(Y = 1, X = 0) + P(Y = 1, X = 1) = 0.08, \\ P(Y = 0) =& P(Y = 0, X = 0) + P(Y = 0, X = 1) = 0.92. \end{aligned}$$

Definition 2.26. Random variables X and Y are **independent** if for all x and y,

$$F(x, y) = F(x)F(y).$$

If X and Y are discrete, this is equivalent to the condition

$$p(x, y) = p_X(x)p_Y(y)$$

for all x and y, and it is also equivalent to the condition

$$p_{Y|X}(y|x) = P_Y(y)$$

for all y and all x such that $P(X = x) > 0$.

Example 2.21. Returning to the previous example, we verify that

$$P(X = 1, Y = 1) \neq P(X = 1)P(Y = 1).$$

Therefore, X and Y are not independent.

2.12 Useful inequalities*

In this section, we introduce some of the most popular inequality in statistics and general mathematics. Interestingly, our probability theories can offer intuitive explanations to these inequalities that are otherwise hard to illustrate.

Theorem 2.10 (Cauchy-Schwarz inequality)**.**

$$\left|\sum x_i y_i\right| \leq \sqrt{\sum x_i^2}\sqrt{\sum y_i^2}$$

Proof. If X, Y have zero means, their correlation can be written as

$$\rho_{XY} = \frac{E(XY)}{\sqrt{E(X^2)E(Y^2)}}$$

Since $|\rho_{XY}| \leq 1$, we always have

$$|E(XY)| \leq \sqrt{E(X^2)E(Y^2)}.$$

Consider $\{x_i\}$ and $\{y_i\}$ as realizations of X and Y with equal probabilities, such that $E(X) = \frac{1}{n}\sum x_i$. The original inequality is thus proved. □

Theorem 2.11 (Jensen's inequality)**.** *For a convex function f, we have*

$$\frac{1}{n}\sum f(x_i) \geq f\left(\frac{1}{n}\sum x_i\right);$$

If f is concave, then

$$\frac{1}{n}\sum f(x_i) \leq f\left(\frac{1}{n}\sum x_i\right).$$

Proof. This is not a proof, but a special case that helps to understand Jensen's inequality. Since

$$Var(X) = E(X^2) - (E(X))^2 \geq 0$$

We have

$$E(X^2) \geq (E(X))^2.$$

Note that $f(X) = X^2$ is a convex function, and $E(*) = \frac{1}{n}\sum *$, we have shown the first inequality. The concave case is the opposite. □

Theorem 2.12 (Markov inequality)**.**

$$P(|X| \geq aE|X|) \leq \frac{1}{a}$$

That is, the probability of $|X|$ deviating from its mean by a multiple of a must be less than $\frac{1}{a}$.

Proof. Define a random variable

$$I_{|X|\geq a} = \begin{cases} 1 & \text{if } |X| \geq a \\ 0 & \text{if } |X| < a \end{cases}$$

Note that $P(|X| \geq a) = E(I_{|X|\geq a})$. It always holds that

$$a \cdot I_{|X|\geq a} \leq |X|$$

Therefore,

$$E\left[a \cdot I_{|X|\geq a}\right] \leq E|X|$$

Hence,

$$P(|X| \geq a) \leq \frac{E|X|}{a}.$$

□

Theorem 2.13 (Chebyshev inequality)**.**

$$P\left(\left|\frac{X-\mu}{\sigma}\right| > a\right) \leq \frac{1}{a^2}$$

That is, the probability of X deviating from its mean by a times the standard deviation must be less than $\frac{1}{a^2}$.

Proof. This is equivalent to

$$P(|X-\mu| > a) \leq \frac{Var(X)}{a^2}$$

This can be shown by taking squares and apply the Markov inequality:

$$P(|X-\mu| > a) = P((X-\mu)^2 > a^2) \leq \frac{E(X-\mu)^2}{a^2} = \frac{Var(X)}{a^2}.$$

□

Given the mean and standard deviation of a random variable X, we know that $\mu \pm 2\sigma$ captures 75% of its possible values; $\mu \pm 3\sigma$ captures 90% of the possible values.

Chapter 3

Continuous random variables

3.1 Review of calculus

Calculus is a prerequisite to work with continuous distributions. The following chapters assume readers are proficient in calculus. We nonetheless review some basic concepts here as a warm-up. This review is not exhaustive, so please refer to a specific textbook if needed for a more comprehensive understanding.

3.1.1 Differentiation

We define the derivative of a function $f(x)$ to be

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{f(x)}$$

Loosely speaking, a function is continuous if there is no jump in the graph, differentiable if the curve is smooth. Some commonly used derivatives:

$$
\begin{aligned}
\frac{d}{dx}(x^n) &= nx^{n-1} \\
\frac{d}{dx}(e^x) &= e^x \\
\frac{d}{dx}(\ln(x)) &= \frac{1}{x} \\
\frac{d}{dx}(\sin(x)) &= \cos(x) \\
\frac{d}{dx}(\cos(x)) &= -\sin(x) \\
(fg)' &= f'g + fg' \\
\left(\frac{f}{g}\right)' &= \frac{f'g - fg'}{g^2} \\
[f(g(x))]' &= f'(g(x))g'(x)
\end{aligned}
$$

When dealing with limits of the form "$\frac{0}{0}$" or "$\frac{\infty}{\infty}$", the L'Hospital rule is very handy.

$$\lim_{x \to a} \frac{f(x)}{g(x)} = \lim_{x \to a} \frac{f'(x)}{g'(x)}.$$

One important application of derivatives is the Taylor's theorem, which gives the approximation of a function around a given point by polynomials. Assume function f is at least k times differentiable, then

$$f(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots + \frac{f^{(k)}(a)}{k!}(x-a)^k + \cdots$$

which means we can approximate a function arbitrarily well by higher order polynomials. Some commonly used Taylor series (expanding around $a = 0$):

$$
\begin{aligned}
\frac{1}{1-x} &= 1 + x + x^2 + x^3 + \cdots \quad \text{for } |x| < 1 \\
e^x &= 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots \\
\sin x &= x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots \\
\cos x &= 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots \\
\ln(1+x) &= x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots \quad \text{for } |x| < 1 \\
\arctan(x) &= x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \cdots \quad \text{for } |x| \leq 1
\end{aligned}
$$

Taylor series are one of the most amazing results in calculus. For example, in the last formula, if we let $x = 1$:

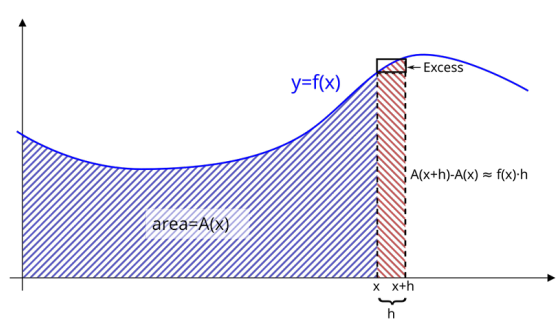
$$\frac{\pi}{4} = \arctan(1) = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots$$

Therefore, we can approximate π by summing up a sequence of fractions:

$$\pi = 4\left(1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots\right).$$

3.1.2 Integration

Integration is the inverse operation of differentiation. Integral has the geometric interpretation as the area under the curve. Let $A(x)$ be the area under the curve of $y = f(x)$. Thus $A(x) = \int_0^x f(t)dt$. The change of the area resulted from a tiny little change of x is approximated by $A(x+h) - A(x) \approx f(x)h$. That is $\frac{A(x+h)-A(x)}{h} = f(x)$. If the change is infinitesimal, $h \to 0$, we have $A'(x) = f(x)$.



The Fundamental Theorem of Calculus: if F is the antiderivative of f, then

$$F(x) = \int_a^x f(t)dt$$

$$\int_a^b f(x)dx = F(b) - F(a)$$

One interpretation of the integral is — the integral of a rate of change of a quantity gives the net change in that quantity. Think about speed and distance: $\int_a^b v(t)dt = s(b) - s(a)$.

Because the integral is just a sum over infinitely many approximating rectangles, $\int_a^b f(x)dx = \lim_{n\to\infty} \sum_{i=1}^n f(x_i)\Delta x$. Integrals behave just like sums. For example, $\frac{1}{b-a}\int_a^b f(x)dx$ has the interpretation of the average of $f(x)$ from a to b.

Indefinite integrals are the general antiderivatives without specifying the interval of the integration. It always comes with a constant C. Some commonly used integrals:

$$\begin{aligned}
\int dx &= x + C \\
\int x^n dx &= \frac{x^{n+1}}{n+1} + C \\
\int e^x dx &= e^x + C \\
\int \frac{1}{x} dx &= \ln|x| + C \\
\int \cos(x)dx &= \sin(x) + C \\
\int \sin(x)dx &= -\cos(x) + C
\end{aligned}$$

Two common integration techniques are *substitution* and *integration by parts*. Integration by substitution applies when substituting part of the integrand makes the integral more approachable.

Example 3.1. Find $\int \sqrt{3x+2}dx$.

Solution: Let $u = 3x + 2$, then $du = 3dx$. Then

$$\int \sqrt{3x+2}dx = \frac{1}{3}\int \sqrt{u}du = \frac{2}{9}u^{3/2} + C = \frac{2}{9}(3x+2)^{3/2} + C.$$

Integration by parts follows the formula: $\int f(x)g'(x)dx = f(x)g(x) - \int f'(x)g(x)dx$.

Example 3.2. Find $\int x \sin x dx$.

Solution: Let $f(x) = x$, $g'(x) = \sin x$. Then $g(x) = -\cos x$. Therefore,

$$\int x \sin x dx = -x\cos x - \int (-\cos x)dx = -x\cos x + \sin x + C.$$

3.2 Amazing integrals*

There are many reasons to learn integrals. But the most compelling reason is that math is no longer the same with integrals. We can have many amazing results with integrals that were otherwise not imaginable. This section is not directly related to our main theme. But let's take a detour just to appreciate the beauty of integrals.

Example 3.3. Show that $\int_{-\infty}^{+\infty} e^{-x^2} dx = \sqrt{\pi}$.

Proof. This is known as Gaussian integral, which is the kernel of the PDF of the normal distribution. It also amazingly relates two of the most famous constants in mathematics. It is not integrable by normal integration techniques. But it can be solved by switching to the polar coordinate.

$$\begin{aligned}
\left(\int_{-\infty}^{+\infty} e^{-x^2} dx\right)^2 &= \int_{-\infty}^{+\infty} e^{-x^2} dx \int_{-\infty}^{+\infty} e^{-y^2} dy \\
&= \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} e^{-(x^2+y^2)} dxdy \\
&= \int_0^{2\pi} \int_0^{\infty} e^{-r^2} r dr d\theta && dA = dxdy = rdrd\theta \\
&= \int_0^{2\pi} \int_0^{\infty} \frac{1}{2} e^{-u} du d\theta && \text{let } u = r^2 \\
&= \frac{1}{2} \int_0^{2\pi} d\theta = \pi.
\end{aligned}$$

□

Example 3.4. Show that $\int_0^{\infty} t^n e^{-t} dt = n!$

Proof. $\Gamma(z) = \int_0^{\infty} t^{z-1} e^{-t} dt$ is known as the Gamma function, which is definitely one of the most interesting functions in mathematics. It is the extension

of factorials to real numbers or even complex numbers. It also has many interesting properties, such as $\Gamma(n) = (n-1)!$, $\Gamma(1/2) = \sqrt{\pi}$, $\Gamma(3/2) = \sqrt{\pi}/2$, $\Gamma'(1) = -\gamma$ and so on. The $(n-1)$ in the Gamma function is due to historical reasons and does not matter in our case. We will prove the integral with n instead of $(n-1)$.

There are many ways to prove this. One is to discover the recursive relationship $\Gamma(n+1) = n\Gamma(n)$. But it does not give a clue why we need this integral to approximate the factorial. We start with an elementary integral

$$\int_0^\infty e^{at} dt = -\frac{1}{a}$$

where $a < 0$. Differentiate both sides n times with respect to a:

$$\begin{aligned}
\int_0^\infty e^{at} t dt &= -(-1)a^{-2} \\
\int_0^\infty e^{at} t^2 dt &= -(-1)(-2)a^{-3} \\
\int_0^\infty e^{at} t^3 dt &= -(-1)(-2)(-3)a^{-4} \\
&\vdots \\
\int_0^\infty e^{at} t^n dt &= (-1)^{n+1} n! a^{-(n+1)}
\end{aligned}$$

Let $a = -1$, we have

$$\int_0^\infty e^t t^n = n!$$

□

Example 3.5. Show that $\int_{-\infty}^{+\infty} \frac{\sin x}{x} dx = \pi$.

Proof. The integrand $\frac{\sin x}{x}$ is also known denoted as $\text{sinc}(x)$, which is widely used in engineering and signal processing. It also has lots of amazing properties, including this integral which evaluates exactly to π. We first transform the integral with Feynman's technique. Define the following integral

$$I(t) = \int_0^\infty e^{-tx} \frac{\sin x}{x} dx$$

Note that $I(0)$ gives back the original integral.

$$I(t) = \int_0^{2\pi} e^{-tx} \frac{\sin x}{x} dx + \int_{2\pi}^{4\pi} e^{-tx} \frac{\sin x}{x} dx + \cdots$$

Focus on a single segment:

$$I_m(t) = \int_{2\pi m}^{2\pi(m+1)} e^{-tx} \frac{\sin x}{x} dx$$

Take derivative with respect to t:

$$I'_m(t) = \int_{2\pi m}^{2\pi(m+1)} e^{-tx}(-x) \frac{\sin x}{x} dx = \int_{2\pi m}^{2\pi(m+1)} e^{-tx}(-\sin x) dx$$

Apply integration by parts twice:

$$\begin{aligned}
\int e^{-tx}(-\sin x)dx =& e^{-tx} \cos x - \int e^{-tx}(-t) \cos x dx \\
=& e^{-tx} \cos x + t \left[e^{-tx} \sin x - \int e^{-tx}(-t) \sin x dx \right] \\
=& e^{-tx} \cos x + t e^{-tx} \sin x + t^2 \int e^{-tx} \sin x dx
\end{aligned}$$

Rearrange,

$$\int e^{-tx}(-\sin x)dx = \frac{1}{1+t^2} \left[e^{-tx} \cos x + t e^{-tx} \sin x \right]$$

Hence,

$$I'_m(t) = \int_{2\pi m}^{2\pi(m+1)} e^{-tx}(-\sin x)dx = \frac{e^{-2\pi t} - 1}{1+t^2} e^{-2\pi m t}$$

Since $\cos(2\pi m) = \cos(2\pi(m+1)) = 1$ and $\sin(2\pi m) = \sin(2\pi(m+1)) = 0$. Let $\alpha = \frac{e^{-2\pi t}-1}{1+t^2}$ which does not depend on m, and $\beta^m = e^{-2\pi m t}$. Then

$$I'_m(t) = \alpha \beta^m$$

Therefore,

$$I'(t) = \sum_{m=0}^{\infty} I'_m(t) = \sum_{m=0}^{\infty} \alpha \beta^m = \frac{\alpha}{1-\beta}$$

Substitute back α and β,

$$I'(t) = \frac{\frac{e^{-2\pi t}-1}{1+t^2}}{1-e^{-2\pi t}} = -\frac{1}{1+t^2}$$

Since $\int \arctan t = \frac{1}{1+t^2} + C$, we have

$$\int_0^\infty I'(t) = I(\infty) - I(0) = [-\arctan t]_0^\infty = -\frac{\pi}{2}$$

where $I(\infty) = 0$ since $e^{-tx} \to 0$ as $t \to \infty$; and $I(0)$ is the original integral. Hence,

$$\int_0^{+\infty} \frac{\sin x}{x} dx = \frac{\pi}{2}.$$

□

3.3 Continuous vs discrete distributions

Most concepts in discrete distributions carry to continuous distributions, by replacing $\sum$ over discrete values to $\int$ over continuous values.

	Discrete	Continuous
PMF/PDF	$P(X = x) = p(x)$	$P(a \le X \le b) = \int_a^b f(x)dx$
CDF	$F(x) = P(X \le x) = \sum_{k \le x} p(k)$	$F(x) = P(X \le x) = \int_{-\infty}^x f(t)dt$
Expectation	$E(x) = \sum_x xP(X = x)$	$E(X) = \int_{-\infty}^{+\infty} xf(x)dx$
LOTUS	$E[g(x)] = \sum_x g(x)P(X = x)$	$E[g(x)] = \int_{-\infty}^{+\infty} g(x)f(x)dx$

3.4 Uniform distribution

Definition 3.1. Let a and b be two given real numbers such that $a < b$. Let X be a random variable such that it is known that $a \le X \le b$ and, for every

subinterval of $[a, b]$, the probability that X will belong to that subinterval is proportional to the length of that subinterval. We then say that the random variable X has the **Uniform distribution** on the interval $[a, b]$. The PDF of X is

$$f(x) = \begin{cases} \frac{1}{b-a} & \text{for } a \leq x \leq b \\ 0 & \text{otherwise} \end{cases}$$

This is a valid PDF since

$$\int_{-\infty}^{+\infty} f(x)dx = \int_a^b \frac{1}{b-a}dx = \frac{1}{b-a}\int_a^b dx = 1.$$

The CDF of X is

$$F(x) = \int_{-\infty}^{x} f(t)dt = \int_a^x f(t)dt = \begin{cases} 0 & x < a \\ \frac{x-a}{b-a} & a \leq x \leq b \\ 1 & x > b \end{cases}.$$

The expectation of X:

$$E(X) = \int_a^b x\frac{1}{b-a}dx = \frac{1}{b-a}\left[\frac{x^2}{2}\right]_a^b = \frac{a+b}{2}.$$

To figure out the variance, first compute

$$E(X^2) = \int_a^b x^2\frac{1}{b-a}dx = \frac{1}{b-a}\left[\frac{x^3}{3}\right]_a^b = \frac{a^2+ab+b^2}{3}$$

Thus,

$$Var(X) = E(X^2) - E^2(X) = \frac{a^2+ab+b^2}{3} - \frac{(a+b)^2}{4} = \frac{(b-a)^2}{12}.$$

Exercise 3.1. Let $X \sim \text{Unif}(0, 1)$. Find $E(X)$ and $Var(X)$.

3.5 Normal distribution

The most widely used model for random variables with continuous distributions is the family of normal distributions. One reason is that many real world samples

appears to be normally distributed (the mass centered around the mean). The other reason is because of the Central Limit Theorem (will be discussed in later chapters), which essentially says the sum (or mean) or any random samples are approximately normal.

Definition 3.2. A random variable Z has the **standard Normal distribution** with mean 0 and variance 1, denoted as $Z \sim N(0,1)$, if Z has a PDF that follows

$$f(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}.$$

This is a valid PDF because $\int_{-\infty}^{\infty} f(z)dz = 1$, which directly follows from Example 3.3. We further verify its mean and variance:

$$E(Z) = \int_{-\infty}^{+\infty} z \cdot \frac{1}{\sqrt{2\pi}} e^{-z^2/2} dz = 0 \quad \text{by symmetry.}$$

$$\begin{aligned}
Var(Z) &= E(Z^2) - (EZ)^2 = E(Z^2) \\
&= \int_{-\infty}^{+\infty} z^2 \cdot \frac{1}{\sqrt{2\pi}} e^{-z^2/2} dz \\
&= \frac{2}{\sqrt{2\pi}} \int_0^{\infty} \underbrace{z}_{u} \cdot \underbrace{z e^{-z^2/2} dz}_{dv} \\
&= \frac{2}{\sqrt{2\pi}} \left\{ \left[z(-e^{-z^2/2}) \right]_0^{\infty} + \underbrace{\int_0^{\infty} e^{-z^2/2} dz}_{\sqrt{2\pi}/2} \right\} \\
&= 1.
\end{aligned}$$

Definition 3.3. The CDF of standard normal distribution is usually denoted by Φ. Therefore,

$$\Phi(z) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{z} e^{-t^2/2} dt.$$

By symmetry, we have $\Phi(-z) = 1 - \Phi(z)$.

Definition 3.4. Let $X = \mu + \sigma Z$ where $Z \sim N(0,1)$. Then we say X has the

Normal distribution with mean μ and variance σ^2, denoted as $X \sim N(\mu, \sigma^2)$. The PDF of X is given by

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right].$$

The mean and variance of X can be easily verified by the properties of expectation and variance.

$$E(X) = E(\mu + \sigma Z) = \mu + \sigma E(Z) = \mu,$$
$$Var(X) = Var(\mu + \sigma Z) = \sigma^2 Var(Z) = \sigma^2.$$

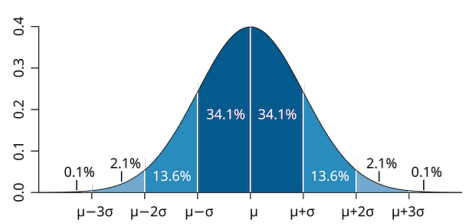
To verify the PDF, we utilize the standard normal CDF:

$$P(X \leq x) = P\left(\frac{X-\mu}{\sigma} \leq \frac{x-\mu}{\sigma}\right) = \Phi\left(\frac{x-\mu}{\sigma}\right)$$

The PDF is the derivative of the CDF,

$$f(x) = \frac{1}{\sigma}\Phi'\left(\frac{x-\mu}{\sigma}\right) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right].$$

The shape of the normal distribution is the famous bell-shaped curve.



The normal distribution has the "three-sigma rule":

$$P(|X-\mu| \leq \sigma) \approx 0.68$$
$$P(|X-\mu| \leq 2\sigma) \approx 0.95$$
$$P(|X-\mu| \leq 3\sigma) \approx 0.997$$

Theorem 3.1. *Let X have the Normal distribution with mean μ and variance σ^2. Let F be the CDF of X. Then the* ***standardization*** *of X*

$$Z = \frac{X - \mu}{\sigma}$$

has the standard normal distribution, and, for all x

$$F(x) = \Phi\left(\frac{x - \mu}{\sigma}\right).$$

To find the value of $\Phi(z)$, we need to use the normal probability table or statistical softwares.

Exercise 3.2. Suppose X has the normal distribution with mean 5 and standard deviation 2. Determine the value of $P(1 < X < 8)$.

Theorem 3.2. *Suppose $X \sim N(\mu, \sigma^2)$. If $Y = aX + b$, then Y has the Normal distribution $Y \sim N(a\mu + b, a^2\sigma^2)$.*

Theorem 3.3. *If the random variables $X_1, \ldots, X_k$ are independent and $X_i \sim N(\mu_i, \sigma_i^2)$. Then*

$$X_1 + \cdots + X_k \sim N(\mu_1 + \cdots + \mu_k, \sigma_1^2 + \cdots + \sigma_k^2).$$

Example 3.6. Suppose the heights (in inches) of women and men independently follow the normal distribution, $W \sim N(65, 1)$, $M \sim N(68, 9)$. (65 inches $\approx$ 165 cm; 68 inches $\approx$ 172 cm) Determine the probability that a randomly selected woman will be taller than a man.

Solution: Let $Z = W - M \sim N(65 - 68, 1 + 9)$. Then $Z \sim N(-3, 10)$. Therefore,

$$P(Z > 0) = P\left(\frac{Z - (-3)}{\sqrt{10}} > \frac{3}{\sqrt{10}}\right) = 1 - \Phi(0.949) = 0.171.$$

Example 3.7 (Distribution of sample mean)**.** Let X_i be the height of a random individual. Assume $X_i \sim N(\mu, \sigma^2)$. Let $\{X_1, X_2, \ldots, X_n\}$ be a sample of n people. The sample mean is calculated as $\overline{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i$. Determine the mean and variance of $\overline{X}_n$.

Solution: By the theorem above, the sum of a series of normal distributions is also normal:

$$\sum_{i=1}^{n} X_i = nX_i \sim N(n\mu, n\sigma^2)$$

since we assume all X_i follow the same distribution. Therefore, the distribution of the sample mean is

$$\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i \sim N(\mu, \sigma^2/n).$$

That is, $\bar{X}_n$ has the normal distribution with mean μ and variance σ^2/n.

How do we understand the sample mean is also a random variable? A sample is a collection of random variables (each observation is a random variable in the sense that the outcome is uncertain). If you were to choose another sample, you would have a different sample mean. Therefore, the sample mean is also a random variable.

3.6 Chi-Square and Student-t

We now introduce two distributions that are closely related to the Normal distribution.

Definition 3.5. Let $V = Z_1^2 + \cdots + Z_n^2$ where $Z_1, Z_2, \ldots, Z_n$ are i.i.d $N(0,1)$. Then V is said to have the **Chi-Square distribution** with n degrees of freedom, denoted as $V \sim \chi^2(n)$.

The χ^2 distribution is a special case of the Gamma distribution that will be introduced in the following sections. In fact, $\chi^2(1)$ is Gamma$(\frac{1}{2}, \frac{1}{2})$; $\chi^2(n)$ is Gamma$(\frac{n}{2}, \frac{1}{2})$.

Definition 3.6. Let

$$T = \frac{Z}{\sqrt{V/n}}$$

where $Z \sim N(0,1)$, $V \sim \chi^2(n)$, and Z is independent of V. Then T is said to have the **Student-t distribution** with n degrees of freedom, denoted as $T \sim t_n$.

Student-t distribution is symmetric and has the similar bell-shaped curve of the Normal distribution but with heavier tail. As $n \to \infty$, t_n distribution approaches the standard Normal distribution.

3.7 Exponential distribution

The Exponential distribution is the continuous counterpart to the Geometric distribution. Recall that a Geometric random variable counts the number of failures before the first success in a sequence of Bernoulli trials. The story of the Exponential distribution is analogous, but we are now waiting for a success in continuous time, where successes arrive at a rate of λ successes per unit of time.

Suppose the number of emails I get in a time interval t is subject to Poisson distribution: $N \sim \text{Pois}(\lambda t)$. Find the PDF of T, the time of waiting until the first email. The waiting time being t is equivalent to $N = 0$ for time period t.

$$P(T > t) = P(N_t = 0) = e^{-\lambda t}\frac{(\lambda t)^0}{0!} = e^{-\lambda t}$$

where $N_t = \#$ emails in $[0, t]$. The CDF of T is

$$F(t) = 1 - P(T > t) = 1 - e^{-\lambda t}.$$

The PDF of T is

$$f(t) = F'(t) = \lambda e^{-\lambda t}.$$

Definition 3.7. A random variable X is said to have the **Exponential distribution** with parameter λ if its PDF is

$$f(x) = \lambda e^{-\lambda x}, \qquad x > 0.$$

We denote this as $X \sim \text{Expo}(\lambda)$.

The corresponding CDF is

$$F(x) = 1 - e^{-\lambda x}, \qquad x > 0.$$

We can standardize the exponential distribution. Let $Y = \lambda X$, then $Y \sim \text{Expo}(1)$. Since $P(Y \leq y) = P(X \leq y/\lambda) = 1 - e^{-y}$.

We now compute the expectation and variance of the exponential distribution. Let $Y \sim \text{Expo}(1)$, then

$$E(Y) = \int_0^\infty ye^{-y}dy = \left[-ye^{-y}\right]_0^\infty + \int_0^\infty e^{-y}dy = 1;$$
$$Var(Y) = E(Y^2) - (EY)^2 = \int_0^\infty y^2e^{-y}dy - 1 = 1.$$

For $X = Y/\lambda$, we have $E(X) = \frac{1}{\lambda}$, $Var(X) = \frac{1}{\lambda^2}$.

Theorem 3.4 (Memoryless property)**.** *If X has the exponential distribution with parameter λ, and let $t > 0$, $h > 0$, then*

$$P(X \geq t + h | X \geq t) = P(X \geq h).$$

Proof. For $t > 0$ we have

$$P(X \geq t) = \int_t^\infty \lambda e^{-\lambda x}dx = e^{-\lambda t}.$$

Hence for each $t > 0$ and each $h > 0$,

$$P(X \geq t + h | X \geq t) = \frac{P(X \geq t + h)}{P(X \geq t)} = \frac{e^{-\lambda(t+h)}}{e^{-\lambda t}} = e^{-\lambda h} = P(X \geq h).$$

□

What are the implications of the memoryless property? If human lifetimes were Exponential, then conditional on having survived to the age of 80, your remaining lifetime would have the same distribution as that of a newborn baby! Clearly, the memoryless property is not an appropriate description for human lifetimes.

The memoryless property is a very special property of the Exponential distribution. In fact, the Exponential is the only memoryless continuous distribution (with support $(0, \infty)$); and Geometric distribution is the only memoryless discrete distribution (with support $0, 1, \ldots$).

Example 3.8. We try to model the waiting time at a bus station. When any bus arrives, suppose the time until the next bus arrives is an Exponential random variable with mean 10 minutes. You arrive at the bus stop at a random time,

not knowing how long ago the previous bus came. What is the distribution of your waiting time for the next bus? What is the average time that you have to wait? What if you know the previous bus left 10 minutes ago, does that change your expected waiting time?

Solution: Let X be the waiting time and we know it is an Exponential distribution. Since $E(X) = 1/\lambda = 10$, the parameter $\lambda = 1/10$. Thus $X \sim \text{Expo}(0.1)$. By the memoryless property, how much longer the next bus will take to arrive is independent of how long ago the previous bus arrived. The average time you have to wait is always 10 minutes.

3.8 Gamma distribution

The Gamma distribution is a continuous distribution on the positive real line; it is a generalization of the Exponential distribution. While an Exponential RV represents the waiting time for the first success, we shall see that a Gamma RV represents the total waiting time for multiple successes.

Definition 3.8. An random variable X is said to have the **Gamma distribution** with parameters a and λ, $a > 0$ and $\lambda > 0$, if it has the PDF

$$f(x) = \frac{\lambda^a}{\Gamma(a)} x^{a-1} e^{-\lambda x}, \quad x > 0$$

We write $X \sim \text{Gamma}(a, \lambda)$.

Verify this is a valid PDF:

$$\int_0^\infty \frac{1}{\Gamma(a)} (\lambda x)^a e^{-\lambda x} \frac{dx}{x} \overset{u=\lambda x}{=} \frac{1}{\Gamma(a)} \int_0^\infty u^a e^{-u} \frac{du}{u} = \frac{\Gamma(a)}{\Gamma(a)} = 1.$$

Taking $a = 1$, the $\text{Gamma}(1, \lambda)$ PDF is $f(x) = \lambda e^{-\lambda x}$, which is the same as $\text{Expo}(\lambda)$. So Exponential distribution is a special case of Gamma distribution.

Let's find the expectation and variance of the Gamma distribution. Let $Y \sim \text{Gamma}(a, 1)$. Recall Γ function has the property $\Gamma(a+1) = a\Gamma(a)$.

$$E(Y) = \int_0^\infty y \cdot \frac{1}{\Gamma(a)} y^{a-1} e^{-y} dy = \frac{1}{\Gamma(a)} \int_0^\infty y^a e^{-y} dy = \frac{\Gamma(a+1)}{\Gamma(a)} = a.$$

Apply LOTUS to evaluate the second moment:

$$E(Y^2) = \int_0^\infty y^2 \cdot \frac{1}{\Gamma(a)} y^{a-1} e^{-y} dy = \frac{1}{\Gamma(a)} \int_0^\infty y^{a+1} e^{-y} dy = \frac{\Gamma(a+2)}{\Gamma(a)} = (a+1)a.$$

Therefore,

$$Var(Y) = (a+1)a - a^2 = a.$$

So for $Y \sim \text{Gamma}(a, 1)$, $E(Y) = Var(Y) = a$. For the general case $X \sim \text{Gamma}(a, \lambda)$, we now show that $X = \frac{Y}{\lambda}$. Note that

$$\begin{aligned} F_X(x) &= P(X \leq x) = P(Y \leq x/\lambda) = F_Y(x/\lambda) \\ f_X(x) &= \frac{dF_X}{dx} = \frac{\partial F_Y}{\partial y} \frac{dy}{dx} = f_Y(y)\lambda \end{aligned}$$

Therefore,

$$f_X(x) = \frac{1}{\Gamma(a)} y^{a-1} e^{-y} \lambda = \frac{\lambda^a}{\Gamma(a)} x^{a-1} e^{-\lambda x}.$$

Hence, we have $E(X) = \frac{a}{\lambda}$, $Var(X) = \frac{a}{\lambda^2}$.

Theorem 3.5. *Let $X_1, \ldots, X_n$ be independent and identical Expo(λ). Then*

$$X_1 + \cdots + X_n \sim Gamma(n, \lambda).$$

Proof. Let's prove by showing the MGFs are equivalent.

$$M_X(t) = E(e^{tX}) = \int_0^\infty e^{tx} \lambda e^{-\lambda x} dx = \frac{\lambda}{\lambda - t} \quad \text{for } t < \lambda$$

Thus, the MGF of $Y = X_1 + \cdots + X_n$ is $M_Y(t) = (M_X(t))^n = \left(\frac{\lambda}{\lambda - t}\right)^n$. We verify this is the MGF of a Gamma distribution. Suppose $Y \sim \text{Gamma}(n, \lambda)$, it has MGF:

$$
\begin{aligned}
M_Y(t) = E(e^{tY}) &= \int_0^\infty e^{ty} \frac{\lambda^n}{\Gamma(a)} y^{n-1} e^{-\lambda y} dy \\
&= \frac{\lambda^n}{(\lambda - t)^n} \int_0^\infty \frac{1}{\Gamma(a)} ((\lambda - t)y)^{n-1} e^{-(\lambda - t)y} (\lambda - t) dy \\
&= \frac{\lambda^n}{(\lambda - t)^n} \int_0^\infty \frac{1}{\Gamma(a)} u^{n-1} e^{-u} du \qquad u = (\lambda - t)y \\
&= \left(\frac{\lambda}{\lambda - t} \right)^n .
\end{aligned}
$$

□

Thus, if X_i represents the $i.i.d$ inter-arrival time. Y has the interpretation of the arrival time until the n-th event.

$$
Y = \sum_{i=1}^{n} X_i = \sum_{i=1}^{n} (\text{time of the i-th arrival}) \sim \text{Gamma}(n, \lambda).
$$

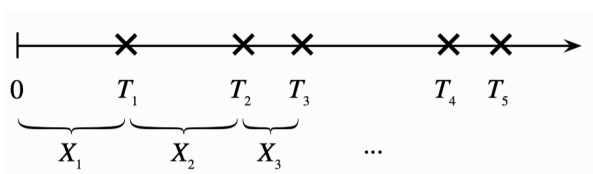


Figure 3.1: Poisson process. The inter-arrival times in a Poisson process are Exponential RVs, while the raw arrival times are Gamma RVs.

Example 3.9 (Service time in a queue)**.** Customer i must wait time X_i for service once reaching the head of the queue. The average service rate is 1 customer per 10 minutes. Assume the service for each customer is independent. If you are the 5th in the queue. What is the expected waiting to be served?

Solution: $X_i \sim$ Expo(0.1). Then $E(X_i) = 10$. Let Y be the time until you are served. Then $Y \sim$ Gamma$(5, 0.1)$. Thus, $E(Y) = \frac{5}{0.1} = 50$ minutes. The probabilities of some selected values:

$$
P(Y = t) = \begin{cases} 0.009 & t = 20 \\ 0.020 & t = 40 \\ 0.009 & t = 70 \end{cases} .
$$

3.9 Beta distribution

The Beta distribution is a continuous distribution on the interval $(0, 1)$. It is a generalization of the Unif$(0, 1)$ distribution, allowing the PDF to be non-constant on $(0, 1)$.

Definition 3.9. A random variable X is said to have the **Beta distribution** with parameters a and b, $a > 0$ and $b > 0$, if its PDF is

$$f(x) = \frac{1}{\beta(a, b)} x^{a-1}(1 - x)^{b-1}, \quad 0 < x < 1$$

where the constant $\beta(a, b)$ is chosen to make the PDF integrate to 1. We write this as $X \sim \text{Beta}(a, b)$.

The Beta distribution takes different shapes for different a and b values. Here are some general patterns:

- If $a = b = 1$, the Beta$(1, 1)$ PDF is constant on $(0, 1)$, equivalent to Unif$(0, 1)$.
- If $a < 1$ and $b < 1$, the PDF is U-shaped and opens upward. If $a > 1$ and $b > 1$, the PDF opens downward.
- If $a = b$, the PDF is symmetric about $1/2$. If $a > b$, the PDF favors values larger than $1/2$. If $a < b$, the PDF favors values smaller than $1/2$.

To make the PDF integrates to 1, the constant $\beta(a, b)$ has to satisfy

$$\beta(a, b) = \int_0^1 x^{a-1}(1 - x)^{b-1} dx.$$

We now try to find this integral:

$$
\begin{aligned}
\beta(a,b) &= \int_0^1 \underbrace{x^{a-1}}_{f} \underbrace{(1-x)^{b-1}}_{g'} dx \\
&= \left[-x^{a-1} \frac{(1-x)^b}{b} \right]_0^1 + \int_0^1 (a-1)x^{a-2} \frac{(1-x)^b}{b} dx \\
&= \frac{a-1}{b} \beta(a-1, b+1) \\
&= \frac{a-1}{b} \cdot \frac{a-2}{b+1} \beta(a-2, b+2) \\
&= \frac{a-1}{b} \cdot \frac{a-2}{b+1} \cdot \frac{a-3}{b+2} \beta(a-3, b+3) \\
&\vdots \\
&= \frac{(a-1)!}{b(b+1)(b+2)\cdots(b+a-2)} \underbrace{\beta(1, a+b-1)}_{\frac{1}{a+b-1}} \\
&= \frac{(a-1)!}{\frac{(b+a-2)!}{(b-1)!}} \cdot \frac{1}{a+b-1} \\
&= \frac{(a-1)!(b-1)!}{(a+b-1)!} \\
&= \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}.
\end{aligned}
$$

Beta distributions are often used as *priors* for parameters in Bayesian inference. We do not cover Bayesian inference in this book. Nonetheless we illustrate this with an example.

Example 3.10 (Beta-Binomial conjugacy)**.** We have a coin that lands Heads with probability p, but we don't know what p is. Our goal is to infer the value of p after observing the outcomes of n tosses of the coin. The larger that n is, the more accurately we should be able to estimate p.

Solution: We model the unknown parameter p as a Beta distribution, $p \sim \text{Beta}(a, b)$. Since we are completely ignorant about this p, we can also model it as the uniform distribution. But we will see that using the Beta distribution is even simpler than the uniform distribution. Let X be the number of heads in n tosses of the coin. Then

$$X|p \sim \text{Bin}(n, p)$$

Apply the Bayes' rule to inverse the conditioning:

$$\begin{aligned} f(p|X=k) &= \frac{P(X=k|p)f(p)}{P(X=k)} \\ &= \frac{\binom{n}{k}p^k(1-p)^{n-k} \cdot \frac{1}{\beta(a,b)}p^{a-1}(1-p)^{b-1}}{\int_0^1 \binom{n}{k}p^k(1-p)^{n-k}f(p)dp} \\ &\propto p^{a+k-1}(1-p)^{b+n-k-1} \end{aligned}$$

This the kernel of $\text{Beta}(a+k, b+n-k)$. The rest is just a normalizing constant. Therefore,

$$p|X=k \sim \text{Beta}(a+k, b+n-k).$$

The *posterior* distribution of p after observing $X=k$ is still a Beta distribution! This is a special relationship between the Beta and Binomial distributions called *conjugacy*: if we have a Beta prior distribution on p and data that are conditionally Binomial given p, then when going from prior to posterior, we don't leave the family of Beta distributions. We say that the *Beta is the conjugate prior of the Binomial.*

3.10 Continuous joint distributions

In order for X and Y to have a continuous joint distribution, we require that the **joint CDF**

$$F_{XY}(x,y) = P(X \leq x, Y \leq y)$$

be *differentiable* with respect to x and y. The partial derivative with respect to x and y is called the joint PDF.

Definition 3.10. If X and Y are continuous with joint CDF F_{XY}, their **joint PDF** is

$$f_{XY}(x,y) = \frac{\partial^2}{\partial x \partial y} F_{XY}(x,y)$$

with requirements $f_{XY}(x,y) \geq 0$ and $\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f_{XY}(x,y)dxdy = 1$.

Definition 3.11. For continuous RVs X and Y, the **marginal PDF** of X is

$$f_X(x) = \int_{-\infty}^{\infty} f_{XY}(x,y)dy.$$

Definition 3.12. For continuous RVs X and Y, the **conditional PDF** of Y given $X = x$ is

$$f_{Y|X}(y|x) = \frac{f_{XY}(x,y)}{f_X(x)}.$$

Theorem 3.6. *For continuous RVs X and Y, we have the Bayes' rule*

$$f_{Y|X}(y|x) = \frac{f_{X|Y}(x|y)f_Y(y)}{f_X(x)},$$

and the Law of total probability

$$f_X(x) = \int_{-\infty}^{\infty} f_{X|Y}(x|y)f_Y(y)dy.$$

Definition 3.13. Random variables X and Y are ***independent*** if for all x and y,

$$F_{XY}(x,y) = F_X(x)F_Y(y)$$

This is equivalent to the condition

$$f_{XY}(x,y) = f_X(x)f_Y(y)$$

for all x and y, and it is also equivalent to the condition

$$f_{Y|X}(y|x) = f_Y(y)$$

for all y and all x such that $f_X(x) > 0$.

Theorem 3.7 (2D LOTUS)**.** *If X and Y are continuous with joint PDF f_{XY}, then*

$$E(g(X,Y)) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} g(x,y)f_{XY}(x,y)dxdy.$$

Example 3.11. Suppose X and Y are uniformly distributed on a unit square $\{(x, y) : x, y \in [0, 1]\}$ with joint PMF

$$f(x, y) = \begin{cases} c & 0 \leq x \leq 1, 0 \leq y \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

Find the value of c. Also find the marginal distribution of X and Y. Are they independent?

Solution: $c = \frac{1}{\text{Area}} = 1$. The marginal distributions are:

$$f_X(x) = \int_0^1 f(x, y)dy = 1$$
$$f_Y(y) = \int_0^1 f(x, y)dx = 1$$

Since $f_X(x)f_Y(y) = f(x, y)$ for all x and y. X, Y are independent.

Example 3.12. Suppose X and Y are uniformly distributed on a disk $\{(x, y) : x^2 + y^2 \leq 1\}$. Find the joint PDF, marginal distributions and conditional distributions. Are X and Y independent?

Solution: The area of the disk is π, therefore

$$f(x, y) = \begin{cases} \frac{1}{\pi} & x^2 + y^2 \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

The marginal distributions are

$$f_X(x) = \int_{-\sqrt{1-x^2}}^{\sqrt{1+x^2}} \frac{1}{\pi} dy = \frac{2}{\pi}\sqrt{1 - x^2}, \qquad -1 \leq x \leq 1$$
$$f_Y(y) = \int_{-\sqrt{1-y^2}}^{\sqrt{1+y^2}} \frac{1}{\pi} dx = \frac{2}{\pi}\sqrt{1 - y^2}, \qquad -1 \leq y \leq 1$$

The conditional distributions are

$$f_{Y|X}(y|x) = \frac{f(x, y)}{f_X(x)} = \frac{\frac{1}{\pi}}{\frac{2}{\pi}\sqrt{1 - x^2}} = \frac{1}{2\sqrt{1 - x^2}}$$

Therefore, $Y|X \sim \text{Unif}(-\sqrt{1-x^2}, \sqrt{1-x^2})$.

Since $f(x,y) \neq f_X(x)f_Y(y)$, X and Y are not independent. This is because knowing the value of X constrains the value of Y.

Example 3.13 (Expected distance between two Uniforms)**.** For $X, Y \overset{iid}{\sim} \text{Unif}(0,1)$, find $E(|X-Y|)$.

Solution:

$$\begin{aligned}
E(|X-Y|) &= \int_0^1 \int_0^1 |x-y| dxdy \\
&= \int_0^1 \int_y^1 (x-y) dxdy + \int_0^1 \int_0^y (y-x) dxdy \\
&= 2\int_0^1 \int_y^1 (x-y) dxdy \\
&= \frac{1}{3}.
\end{aligned}$$

Chapter 4

Sampling distribution

4.1 Samples and statistics

We model real-world uncertain events with random variables. We have also introduced various distributions suitable to model different kinds of events. However, we never observe the full distribution or the true parameters of the assumed distribution. Instead, we only observe a sample of that random variable. We can only infer the properties of the distribution from a limited sample. For example, suppose we model the hight of an Asian women with a normal distribution. But we never know exactly what the mean and variance are. We can only observe a sample of the distribution.

In statistics, the conceptual distribution F is called the **population distribution**, or just the **population**.[1] It is tempting to think of the population as all the observations (e.g. all the population on the planet), but this is not exactly correct. The population distribution is more of a mathematical abstraction or an assumption. Suppose we are modeling the height of human being, even if we have all the observations on the planet, that does not include the people that have died or yet to be born. Thus, it is still a sample of the assumed distribution.

A collection of random variables $\{X_1, X_2, \ldots, X_n\}$ is a **random sample** from the population F if X_i are **independent and identically distributed (*i.i.d*)**

[1] This section is based on Bruce Hansen's *Probability and Statistics for Economists.*

with distribution F. What we mean by *i.i.d* is that $X_1, \ldots, X_n$ are mutually independent and have exactly the same distribution $X_i \sim F$. Survey sampling is an useful metaphor to understand random sampling, in which we randomly select a subset of the population with equal probability. The **sample size** n is the number of individuals in the sample.

A **data set** is a collection of numbers, typically organized by observation. We sometimes call a data set also as a sample. But it should not be confused with the random sample defined above. As the former is a collection of random variables, whereas the latter is one **realization** of the random variables.

Typically, we will use X without the subscript to denote a random variable or vector with distribution F, X_i with a subscript to denote a random observation in the sample, and x_i or x to denote a specific or realized value.

The problem of **statistical inference** is to learn about the underlying process — the population distribution or data generating process — by examining the observations. In most cases, we assume the population distribution and want to learn about the its parameters (e.g. μ and σ^2 in the normal distribution). As a convention, we use greek letters to denote population parameters.

A **statistic** is a function of the random sample $\{X_1, X_2, \ldots, X_n\}$. Recall that there is a distinction between random variables and their realizations. Similarly there is a distinction between a statistic as a function of a random sample — and is therefore a random variable as well — and a statistic as a function of the realized sample, which is a realized value. When we treat a statistic as random we are viewing it is a function of a sample of random variables. When we treat it as a realized value we are viewing it as a function of a set of realized values. One way of viewing the distinction is to think of "before viewing the data" and "after viewing the data". When we think about a statistic "before viewing" we do not know what value it will take. From our vantage point it is unknown and random. After viewing the data and specifically after computing and viewing the statistic the latter is a specific number and is therefore a realization. It is what it is and it is not changing. The randomness is the process by which the data was generated — and the understanding that if this process were repeated the sample would be different and the specific realization would be therefore different. The distribution of a statistic is called the **sampling distribution**, since it is the distribution induced by sampling.

An **estimator** $\hat{\theta}$ for a population parameter θ is a statistic intended to infer θ.

It is conventional to use the hat notation $\hat{\theta}$ to denote an estimator. Note that $\hat{\theta}$ is a statistic and hence also a random variable. We call $\hat{\theta}$ an **estimate** when it is a specific value (or realized value) calculated in a specific sample.

A standard way to construct an estimator is by the analog principle. The idea is to express the parameter θ as a function of the population F, and then express the estimator $\hat{\theta}$ as the analog function in the sample.

For example, suppose we want to construct an estimator for the population mean $\mu = E(X)$. By definition, if each value of X is of equal probability, μ is simply the average. By analogy, we construct the **sample mean** as $\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i$. It is conventional to denote a sample average by the notation "X bar". Because it is an estimator for μ, we also denote it as $\hat{\mu} = \bar{X}_n$. Note that from different samples we calculate different estimates. In one sample, $\hat{\mu} = 6.5$; in another sample, $\hat{\mu} = 6.7$. All of them are erroneous estimate of the true parameter μ. The question is therefore how close they are to the true parameter. To answer this question, we need to study the distribution of the sample mean.

4.2 Descriptive statistics*

In this section, we familiarize ourselves with data sets. In most of the applications in economics and finance, a data set is just a table. Table 4.1 shows a snippet from the dataset China Household Finance Survey (CHFS).[2]Each column is like a random variable $X, Y, \ldots$ The data set is like a random sample:

X_1	Y_1	Z_1
X_2	Y_2	Z_2
X_3	Y_3	Z_3
$\vdots$	$\vdots$	$\vdots$

The numbers are realizations of the random sample. Typically, each row represents one observation. In Table 4.1, the unit of observation is household. But it could also be one individual, one city, or one period in time.

[2]The China Household Finance Survey is conducted by the China Family Finance Survey and Research Center in the Southwestern University of Finance and Economics. The data cited is from the survey conducted in 2019.

	hhid	total_income	total_asset	...
1	2019007329	27,235	13,000	
2	2017032986	14,562	85,500	
3	2019022586	563,294	9,623,694	
4	2019018619	19,547	403,540	
5	2017014003	122,125	737,547	
6	2013023600	151,083	687,526	
7	2017015493	70,065	800,547	
8	2013024998	67,376	408,350	
9	2015024693	19,841	29,188	
	⋮			

Table 4.1: A snippet from CHFS (2019)

The data contained in a data set can be categorical or numerical. **Categorical data** represents characteristics or attributes that can be divided into distinct groups or categories. These categories are typically qualitative and can be either nominal or ordinal.

- **Nominal Data:** Categories without a specific order.
  - Examples: Colors (red, blue, green), types of fruit (apple, banana, orange).
- **Ordinal Data:** Categories with a meaningful order.
  - Examples: Movie ratings (poor, fair, good, excellent), education levels (high school, bachelor's, master's).

Categorical data is commonly visualized using bar charts or pie charts.

Numerical data represents quantities and can be measured and ordered. It is quantitative and can be either discrete or continuous.

- **Discrete Data:** Countable values, often integers.
  - Examples: Number of students in a class, number of cars in a carpark.
- **Continuous Data:** Data that can take any value within a range.
  - Examples: Height, weight, temperature.

Numerical data is often analyzed using statistical measures like mean, median, and standard deviation. As we have seen before, the sample mean is defined as

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i;$$

and the sample variance is:

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2.$$

We will explain later why we need $(n-1)$ instead of n. Sample mean and variance tells us where the data is centered and how dispersed they are. Sometimes, we are also interested in the range of the data. We often use quartiles and percentiles to get a rough ideal of the distribution.

Quartiles divide a dataset into four equal parts:

- 1st Quartile (Q1): The median of the lower half (25%) of the dataset.
- 2nd Quartile (Q2): The median (50%) of the dataset.
- 3rd Quartile (Q3): The median of the upper half (75%) of the dataset.

Percentiles divide a dataset into 100 equal parts. They indicate the value below which a given percentage of observations fall:

- Example: The 90th percentile is the value below which 90% of the data points lie.

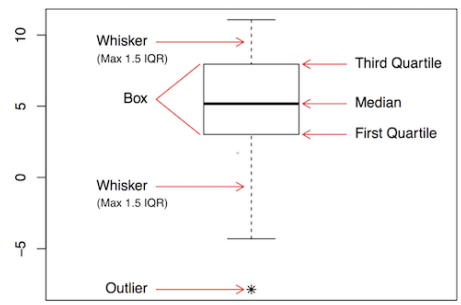
The quartiles of the CHFS is data is summarized below:

Variable	N	Missing	Mean	SD	P25	P50	P75
hhwage_inc	107,008	36,456	76,828	103,227	27,727	54,000	96,120

N is the total number of observations. *Missing* is the number of missing observations. It is always recommended to check the missing observations. If the values are missing randomly, it is usually not a big problem. But if they are missing

due to particular reasons, it could potentially violate the $i.i.d$ assumption of the sample.

Histogram is one common way to visualize the distribution. We have seen it many times. Boxplot is another useful visualization:



If an observation is too far away from the mass of the data. We might consider it as an *outlier*. An outlier may be due to measurement errors or experimental errors. If this is the case, the outlier should be excluded from the dataset. But we should be very careful to do so. Because it is hard to tell if an outlier is an error or an unusual value.

4.3 Law of large numbers

We now introduce two important theorems describing the behavior of the sample mean as the sample size grows. Throughout this section and the next, we assume $X_1, X_2, \ldots, X_n$ are i.i.d RVs drawn from a population with mean μ and variance σ^2. The sample mean is defined as

$$\bar{X}_n = \frac{X_1 + \cdots + X_n}{n}.$$

As we have discussed previously, the sample mean is itself a random variable with mean and variance:

$$E(\bar{X}_n) = \frac{1}{n}E(X_1 + \cdots + X_n) = \frac{1}{n}(E(X_1) + \cdots + E(X_n)) = \mu,$$
$$Var(\bar{X}_n) = \frac{1}{n^2}Var(X_1 + \cdots + X_n) \overset{iid}{=} \frac{1}{n^2}(Var(X_1) + \cdots + Var(X_n)) = \frac{\sigma^2}{n}.$$

The law of large numbers (LLN) says that as n grows, the sample mean $\bar{X}_n$ converges to the true mean μ.

Theorem 4.1 (Strong law of large numbers)**.** *The sample mean $\bar{X}_n$ converges to the true mean μ point-wise as $n \to \infty$, with probability* 1*. In other words, the event $\bar{X}_n \to \mu$ has probability* 1*.*

Theorem 4.2 (Weak law of large numbers)**.** *For all $\epsilon > 0$, $P(|\bar{X}_n - \mu| > \epsilon) \to 0$ as $n \to \infty$. (this is known as converge in probability).*

We don't need a rigorous proof here. But an intuitive proof is obvious. As $n \to \infty$, $Var(\bar{X}_n) = \frac{\sigma^2}{n} \to 0$. The random variable $\bar{X}_n$ becomes fixed at μ as n becomes large. Thus, it converges to μ in a probabilistic sense.

It seems that the LLN just states the obvious. But it has wide applications in daily time that you might not even realize. What it says is essentially this: the uncertainty at the individual level becomes certain when aggregating together; the risks that are unmanageable at the individual level becomes manageable collectively. Think about a rare disease, it happens at 1 out of a million probability. For each individual, no one knows if they will get the disease or not. But as the sample size gets large, suppose we have one billion population, the LLN says the sample mean will be very close the true mean. That is, there will be almost surely 1000 people being infected by the disease. We provide two more examples.

Example 4.1 (Lottery)**.** A lottery company is designing a game with a 6-digit format. Each time someone buys a ticket, they receive a randomly generated 6-digit number. Only one number will win the grand prize of 10 million dollars. What should the company charge per ticket to break even?

Solution: The probability of winning the game is $p = 1/10^6$. Suppose the company has sold n tickets. The price for each ticket is x. The revenue for the company is therefore xn. By the LLN, the cost of the company should be very close to $10^7 np$. The break even point is $xn = 10^7 np$. So $x = 10^7 p = 10$. Therefore, if the company sells each ticket above 10 dollars. The business is surely profitable as long as n is large. If the company is a monopoly, it can reap as much profit as it desires as long as they know the basic probability theory! The same can be said about gambling companies.

Example 4.2 (Insurance)**.** Insurance is anther great application of the LLN. It is essentially the same as the the lottery game but most people do not realize it.

Suppose there is a disease with infection rate of 1 out of 1 million. The medical expenditure to cure the disease is 10 million dollars. How much the insurance company should charge per customer to cover this disease?

Solution: The solution is essentially the same as above. Suppose the premium for the insurance product is x. The revenue of the company by selling the premium is xn. The cost is — when one customer is infected, the company has to pay the medical cost —$10^7 np$. The break even price for the insurance premium is thus 10 dollars.

What is the implication of this insurance? Without the insurance, each individual either chooses to set aside 10 million dollars pre-cautiously for the disease (if he is rich enough) or be exposed to the risk completely uncovered. The insurance product enables everyone to get covered at a cost of just 10 dollars. It is a typical example that the unmanageable risk at the individual level becomes manageable collectively.

4.4 Central limit theorem

The LLN shows the convergence of the sample mean to the population mean. What about the entire sample distribution? This is addressed by the central limit theorem (CLT), which, as its name suggests, is a limit theorem of **central importance** in statistics.

The CLT states that for large n, the distribution of $\bar{X}_n$ after standardization approaches a standard Normal distribution, regardless of the underlying distribution of X_i. By **standardization**, we mean that we subtract μ, the expected value of $\bar{X}_n$, and divide by $\sigma/\sqrt{n}$, the standard deviation of $\bar{X}_n$.

Theorem 4.3 (Central limit theorem)**.** *As $n \to \infty$,*

$$\sqrt{n}\left(\frac{\bar{X}_n - \mu}{\sigma}\right) \to N(0,1) \text{ in distribution.}$$

In other words, the CDF of the left-hand side approaches the CDF of the standard normal distribution.

Proof. We will prove the CLT assuming the MGF of the X_i exists, though the theorem holds under much weaker conditions. Without loss of generality let

$\mu = 1, \sigma^2 = 1$ (since we standardize it anyway). We show that the MGF of $\sqrt{n}\bar{X}_n = (X_1 + \cdots + X_n)/\sqrt{n}$ converges to the MGF of the $N(0,1)$.

The MGF of $N(0,1)$ is

$$\begin{aligned}
E(e^{tX}) &= \int_{-\infty}^{\infty} e^{tx} \cdot \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx \\
&= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-x^2/2+tx} dx \\
&= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(x-t)^2+\frac{1}{2}t^2} dx \\
&= e^{\frac{t^2}{2}} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(x-t)^2} dx \\
&= e^{t^2/2}
\end{aligned}$$

Compute the MGF of $\sqrt{n}\bar{X}_n$:

$$\begin{aligned}
E(e^{\sqrt{n}\bar{X}_n}) &= E(e^{t(X_1+\cdots+X_n)/\sqrt{n}}) \\
&= E(e^{tX_1/\sqrt{n}})E(e^{tX_2/\sqrt{n}}) \cdots E(e^{tX_n/\sqrt{n}}) \\
&= \left[E(e^{tX_i/\sqrt{n}})\right]^n \qquad \text{since } i.i.d \\
&= \left[E\left(1 + \frac{tX_i}{\sqrt{n}} + \frac{t^2X_i^2}{2n} + o(n^{-1})\right)\right]^n \\
&= \left[1 + \frac{t}{\sqrt{n}}E(X_i) + \frac{t^2}{2n}E(X_i^2) + o(n^{-1})\right]^n \\
&= \left[1 + \frac{t^2}{2n} + o(n^{-1})\right]^n \\
&= \left[1 + \frac{t^2/2}{n} + o(n^{-1})\right]^n \\
&\to e^{t^2/2} \qquad \text{as } n \to \infty
\end{aligned}$$

Therefore, the MGF of $\sqrt{n}\bar{X}_n$ approaches the MGF of the standard normal. Since MGF determines the distribution, the distribution of $\sqrt{n}\bar{X}_n$ also approaches the standard normal distribution. □

The CLT tells us about the limiting distribution of $\bar{X}_n$ as $n \to \infty$. That means,

we can reasonably approximate the distribution $\bar{X}_n$ with normal distribution when n is a finite large number —

$$\bar{X}_n \approx N(\mu, \sigma^2/n) \quad \text{for large } n.$$

The Central Limit Theorem was first proved by Pierre-Simon Laplace in 1810. Let's take a moment to admire the generality of this result. The distribution of the individual X_i can be anything in the world, as long as the mean and variance are finite. This does mean the distribution of X_i is irrelevant, however. If the distribution is fairly close to normal, the result would hold for smaller n. If the distribution is far away from normal, it would take larger n to converge.

The CLT gives the distribution of the sample mean regardless of the underlying distribution. This allows to assess the "quality" of the sample mean — how close it is to the true mean. The LLN tells us the larger the sample, the closer the sample mean to the population mean. The CLT tells us the distribution of the sample mean for sample size n. For smaller n, the distribution is more spread-out (a normal distribution with large σ^2); hence the uncertainty is huge, other values are more likely. For larger n, the uncertainty is reduced, most values would be centered around the true mean. We will delve deeper into this when we get to hypothesis testing.

Example 4.3. Suppose that a fair coin is tossed 900 times. Approximate the probability of obtaining more than 395 heads.

Solution: Let $H = \sum_{i=1}^{900} X_i$ be the number of heads, where $X_i \sim \text{Bern}(\frac{1}{2})$. We could compute the probability by

$$P(H > 495) = \sum_{k=496}^{900} \binom{900}{k} \left(\frac{1}{2}\right)^k \left(\frac{1}{2}\right)^{900-k}$$

But this is quite tedious. Because $n = 900$ is reasonably large, we can apply the CLT:

$$\frac{1}{n}\sum_{i=1}^{900} X_i \sim N(\mu, \sigma^2/n) \quad \text{or}$$
$$\sum_{i=1}^{900} X_i \sim N(n\mu, n\sigma^2)$$

We know $\mu = E(X_i) = \frac{1}{2}$, $\sigma^2 = Var(X_i) = \frac{1}{4}$. Thus $H \sim N(450, 225)$. Therefore,

$$P(H > 495) = 1 - P(H \leq 495) \approx 1 - \Phi\left(\frac{495 - 450}{15}\right) = 0.0013.$$

4.5 Estimator accuracy

This section introduces some measures regarding the accuracy of an estimator.

Definition 4.1. The **bias** of an estimator $\hat{\theta}$ of a parameter θ is

$$\text{Bias}[\hat{\theta}] = E(\hat{\theta}) - \theta.$$

We say that an estimator is **biased** if its sampling is incorrectly centered. We say that an estimator is **unbiased** is the bias is zero.

Theorem 4.4. *$\bar{X}_n$ is **unbiased** for $\mu = E(x)$ if $E(X) < \infty$.*

Proof.

$$E(\bar{X}_n) = E\left(\frac{1}{n}\sum_{i=1}^{n} X_i\right) = \frac{1}{n}\sum_{i=1}^{n} E(X_i) = \frac{1}{n}\sum_{i=1}^{n} \mu = \mu.$$

□

Theorem 4.5. *If $\hat{\theta}$ is an unbiased estimator of θ, then $\hat{\beta} = a\hat{\theta} + b$ is an unbiased estimator of $\beta = a\theta + b$.*

But obtaining an unbiased estimator is not always as straightforward as it seems. Consider the sample variance as an estimator for the population variance. By the analog principle, the sample variance should be

$$\begin{aligned}
\hat{\sigma}^2 &= \frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X_n})^2 \\
&= \frac{1}{n}\sum_{i=1}^{n}(X_i - \mu)^2 - \left(\frac{1}{n}\sum_{i=1}^{n}(X_i - \mu)\right)^2 \\
&= \tilde{\sigma}^2 - (\bar{X}_n - \mu)^2
\end{aligned}$$

We know that

$$E(\tilde{\sigma}^2) = \frac{1}{n}\sum_{i=1}^{n} E(X_i - \mu)^2 = \sigma^2$$

Thus, if we compute the bias of this estimator:

$$\begin{aligned}
E[\hat{\sigma}^2] &= \sigma^2 - \frac{\sigma^2}{n} = \left(1 - \frac{1}{n}\right)\sigma^2 \\
\text{Bias}[\hat{\sigma}^2] &= -\frac{\sigma^2}{n} \neq 0
\end{aligned}$$

Therefore, the estimator $\hat{\sigma}^2$ is a biased estimator for σ^2! To correct the bias, we divide the sample sum of squares by $(n-1)$.

$$s^2 = \frac{n}{n-1}\hat{\sigma}^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X_n})^2.$$

It is straightforward to see that s^2 is an unbiased estimator for σ^2. We call s^2 the **bias-corrected variance estimator**.

Theorem 4.6. *s^2 is an unbiased estimator for σ^2 if $E(X^2) < \infty$.*

Definition 4.2. The mean square error of an estimator $\hat{\theta}$ for θ is

$$\text{MSE}[\hat{\theta}] = E\left[(\hat{\theta} - \theta)^2\right].$$

By expanding the square we find that

$$\begin{aligned}
\text{MSE}[\hat{\theta}] &= E\left[(\hat{\theta}-\theta)^2\right] \\
&= E\left[(\hat{\theta}-E[\hat{\theta}]+E[\hat{\theta}]-\theta)^2\right] \\
&= E\left[(\hat{\theta}-E[\hat{\theta}])^2\right] + 2E(\hat{\theta}-E[\hat{\theta}])(E[\hat{\theta}]-\theta) + (E[\hat{\theta}]-\theta)^2 \\
&= Var[\hat{\theta}] + (\text{Bias}[\hat{\theta}])^2.
\end{aligned}$$

Thus the MSE is the variance plus the squared bias. The MSE as a measure of accuracy combines the variance and bias.

Theorem 4.7. *For any estimator with a finite variance, we have*

$$MSE[\hat{\theta}] = Var[\hat{\theta}] + (Bias[\hat{\theta}])^2.$$

Definition 4.3. An estimator is **consistent** if $\text{MSE}[\hat{\theta}] \to 0$ as $n \to \infty$.

Bias is the property of an estimator for finite samples. Consistency is the property of an estimator when the sample size gets large. It means that for any given data distribution, there is a sample size n sufficiently large such that the estimator $\hat{\theta}$ will be arbitrarily close to the true value θ with high probability. In practice, we usually do not know how large this n has to be. But it is a desirable property for an estimator to be considered a "good" estimator.

For unbiased estimator, MSE is solely determined by the variance of the estimator. Recall that the variance for the sample mean is $Var(\bar{X}_n) = \sigma^2/n$. But this is not a very useful formula because the it depends on unknown parameter σ^2. We need to replace these unknown parameters by estimators. To put the latter in the same units as the parameter estimate we typically take the square root before reporting. We thus arrive at the following concept.

Definition 4.4. A **standard error** of an estimator $\hat{\theta}$ is defined as

$$SE(\hat{\theta}) = \hat{V}^{1/2}$$

where $\hat{V}$ is the estimator for $Var[\hat{\theta}]$.

Definition 4.5. The **standard error** for $\bar{X}_n$ is

$$SE(\bar{X}_n) = \frac{s}{\sqrt{n}}$$

where s is the bias-corrected estimator for σ.

Note the difference between **standard error** and **standard deviation**. Standard deviation describes the dispersion of a distribution. Standard error is the standard deviation of an *estimator*. It indicates the "precision" of the estimator, thereby carrying a sense of "error". The smaller the standard error, the more precise the estimator.

4.6 Confidence intervals

Confidence intervals provide a method of adding more information to an estimator $\hat{\theta}$ when we wish to estimate an unknown parameter θ. We can find an interval (A, B) that we think has high probability of containing θ. The length of such an interval gives us an idea of how closely we can estimate θ.

Definition 4.6. A $100(1-\alpha)\%$ **confidence interval (CI)** for θ is an interval $[L(\theta), U(\theta)]$ such that the probability that the interval contains the true θ is $(1-\alpha)$.

Due to randomness we rarely seek a confidence interval with 100% coverage as this would typically need to be the entire parameter space. Instead we seek an interval which includes the true value with reasonably high probability. Standard choices are $\alpha = 0.05$ and 0.10, corresponding to 95% and 90% confidence.

Confidence intervals are reported to indicate the degree of precision of our estimates. The narrower the confidence interval, the more precise the estimate. Because a small range of values contains the true parameter with high probability.

With the help of the CLT, it is not hard to find the CI for the sample mean $\bar{X}_n$. Let's set $\alpha = 5\%$, that is, we are trying to find the CI that contains the true mean 95% of the times. Assume our sample size n is large enough to invoke the CLT, we thus have

$$\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \sim N(0,1)$$

We want to find L and U such that

$$P(L \leq \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \leq U) = 1 - 2\Phi(L) = 0.95$$

since the normal distribution is symmetric, $L = -U$. By looking at the CDF of standard normal, we get $L = -1.96$, $U = 1.96$. So the interval is

$$-1.96 \leq \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \leq 1.96$$

With a little rearrangement, we have

$$\bar{X}_n - 1.96\frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{X}_n + 1.96\frac{\sigma}{\sqrt{n}}$$

Thus, the interval $\left[\bar{X}_n - 1.96\frac{\sigma}{\sqrt{n}}, \bar{X}_n + 1.96\frac{\sigma}{\sqrt{n}}\right]$ contains the true mean 95% of the times.

Theorem 4.8. *The* $100(1-\alpha)\%$ *confidence interval for the sample mean* $\bar{X}_n$ *is* $\bar{X}_n \pm z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$*, where* $z_{\alpha/2}$ *is the critical value such that* $\Phi(z_{\alpha/2}) = \frac{\alpha}{2}$*.*

In practice, because we do not know $\sigma/\sqrt{n}$, we replace it with the standard error $s/\sqrt{n}$. Thus, we compute the confidence interval as $\bar{X}_n \pm z_{\alpha/2}SE$. However, this replacement is not without risk. When the sample size is small, s is a very poor estimate of σ. For the approximation to be valid, we require either the sample size is large enough ($n \geq 30$ at least) or the population distribution is nearly normal. Some commonly used confidence levels:

- 90% CI: $\alpha = 0.1$, $z_{0.05} = 1.645$
- 95% CI: $\alpha = 0.05$, $z_{0.025} = 1.96$
- 99% CI: $\alpha = 0.01$, $z_{0.005} = 2.58$

We go through some common misunderstandings about confidence intervals through an example.

Example 4.4. A random sample of 50 college students were asked how many exclusive relationships they have been in so far. This sample yielded a mean

of 3.2 and a standard deviation of 1.74. Estimate the true average number of exclusive relationships using this sample. [3]

Solution: We know $\bar{X} = 3.2$, $s = 1.74$. The standard error is

$$SE = \frac{1.74}{\sqrt{50}} \approx 0.5$$

The approximate 95% CI is therefore

$$\bar{X} \pm 1.96SE \approx 3.2 \pm 0.5 = (2.7, 3.7).$$

Now check the following interpretations (true or false):

1. We are 95% confident that the average number of exclusive relationships in this sample is between 2.7 and 3.7.
   - False. The CI definitely contains the sample mean $\bar{X}$.
2. 95% of college students have been in 2.7 to 3.7 exclusive relationships.
   - False. The CI is about covering the population mean, not for covering 95% of the entire population.
3. There is 0.95 probability that the true mean falls in the interval (2.7, 3.7).
   - False. The true mean μ is a fixed number, not a random one that happens with a probability.
4. The interval (2.7, 3.7) has probability of 0.95 of enclosing the true mean number of exclusive relationships of college students.
   - False. The true mean is either in the interval or not. There is no uncertainty involved.
5. If a new random sample of size 50 is taken, we are 95% confident that the new sample mean will be between 2.7 and 3.7.
   - False. The confidence interval is for covering the population mean, not for covering the mean of another sample.

[3] This section and the next are based on Dr. Yibi Huang's lecture slides.

6. This confidence interval is not valid since the number of exclusive relationships is integer-valued. Neither the population nor sample is normally distributed.

   - False. The construction of the CI only uses the normality of the sampling distribution of the sample mean (by the CLT). Neither the population nor the sample is required to be normally distributed.

So what is exactly the thing that has a 95% change to happen? It is the **procedure to construct the 95% interval**. About 95% of the intervals constructed following the procedure will cover the true population mean μ. After taking the sample and an interval is constructed, the constructed interval either covers μ or it doesn't. But if we were able to take many such samples and reconstruct the interval many times, 95% of the intervals will contain the true mean.

4.7 Hypothesis testing

One day I woke up in the morning and came up with a question: Am I the average height of a Chinese man? I hypothesize that I am. My height is $\sqrt{3} \approx 1.73$ meters. Let μ be the average height of Chinese men. My hypothesis is

$$H_0 : \mu = \sqrt{3}$$

This is called the **null hypothesis**. I also suspect that the average Chinese men is taller than me (if the original hypothesis is false).

$$H_1 : \mu > \sqrt{3}$$

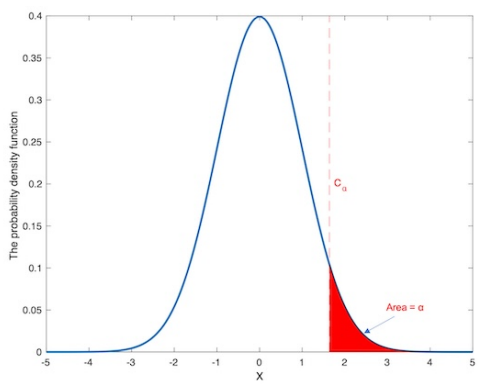
This is called the **alternative hypothesis**. How am I able to test which hypothesis is true? I can answer this question by collecting a small sample. Suppose I asked 50 people around me, and computed a sample average of $\bar{X} = 1.76$ and standard deviation $s = 0.1$. Does it prove or disprove the hypothesis?

By the CLT, we know $\frac{\bar{X}-\mu}{\sigma/\sqrt{n}}$ is approximately standard normal. Suppose H_0 is true, we compute

$$Z = \frac{\bar{X} - \mu}{SE} = \frac{1.76 - \sqrt{3}}{0.1/\sqrt{50}} = 1.98$$

which is know as the z**-score** of the sample mean. Thus, the probability of $\bar{X} > 1.76$ is

$$P(\bar{X} > 1.76) = 1 - \Phi\left(Z = 1.98\right) = 0.023$$



That is the small red region in the graph. That means, if H_0 is true, we only have a very small chance of observing $\bar{X} = 1.76$. Therefore, the hypothesis H_0 is likely false. In other words, if we reject H_0 and accept the alternative H_1, the probability that we have made a mistake is very low.

However, if we had observed $\bar{X} = 1.73$. The probability above is 0.56. That means we are very likely to observe this value if H_0 is true. In this case, it would be reasonable to accept H_0. In other words, we do not have strong enough evidence to reject the hypothesis.

The probability represented by the red area is called the p**-value**. The p-value is the probability of obtaining test results at least as extreme as the result actually observed, under the assumption that the null hypothesis is correct. A very small p-value means that such an extreme observed outcome would be very unlikely under the null hypothesis. Thus, <u>The smaller the p-value, the stronger the evidence against the H_0.</u>

In some studies, we can simply report the p-value and let people judge whether the evidence is strong enough. In other studies, we prefer to select a cut-off value α, call the **significance level**, and follow the rule:

- If the p-value $< \alpha$, reject H_0;
- If the p-value $> \alpha$, do not reject H_0.

Commonly used significance levels: 0.05 and 0.01. And we like to use the word "significant" to describe the test result:

- A test with p-value < 0.05 is said to be (statistically) **significant**;
- A test with p-value < 0.01 is said to be highly **significant**.

When we make a decision about accepting or rejecting a hypothesis, there are chances that we make a mistake. There are two types of mistakes: **Type 1 error** and **Type 2 error**.

		Decision	
		Reject H_0	Fail to reject H_0
Truth	H_0 is true	Type 1 error	✓
	H_0 is false	✓	Type 2 error

Type 1 error is rejecting the H_0 when it is true. **Type 2 error** is failing to reject the H_0 when it is false. Usually, it is more important to control the Type 1 error than the the Type 2 error. That is, we want to minimize the chance of falsely rejecting the null hypothesis.

In the example above, we reject the null hypothesis on the ground that there is only 2.3% of the chance that we could observe this sample. Therefore, the probability of Type 1 error is only 2.3%.

If we make decisions based on a significance level, the significance level is the Type 1 error rate. In other words, when using a 5% significance level, there is 5% chance of making a Type 1 error.

$$P(\text{Type 1 error}|H_0 \text{ is true}) = \alpha$$

This is why we prefer small values of α—smaller α reduces the Type 1 error rate. However, significance level doesn't control Type 2 error rate.

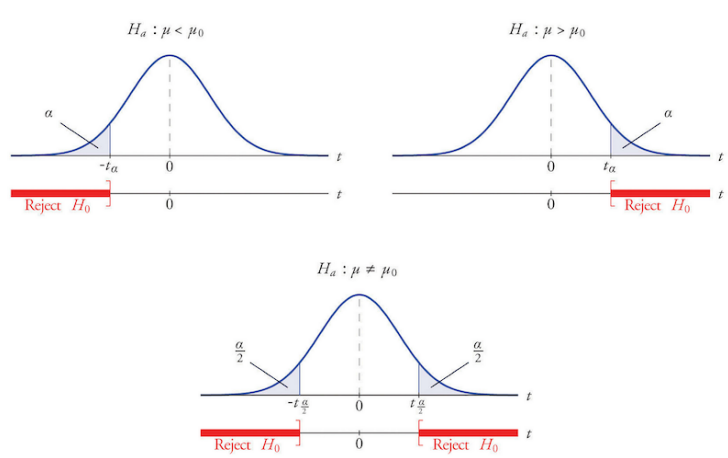
Here we summarize the steps of testing for a population mean:

1. Set up the hypothesis:
   - $H_0 : \mu = \mu_0$
   - $H_1 : \mu <$ or $>$ or $\neq \mu_0$
2. Check assumptions and conditions
   - $i.i.d$
   - Nearly normal distribution or the sample size is large enough
3. Compute the test statistic and the p-value:

$$Z = \frac{\bar{X} - \mu}{SE} \quad \text{where } SE = \frac{s}{\sqrt{n}}$$

4. Make decision:
   - If p-value $< \alpha$, reject H_0
   - If p-value $> \alpha$, do not reject H_0

We notice that the **two-sided** hypothesis tests are very closed related to the concept of confidence intervals. A two-sided test means we are interested in rejection regions on both sides of the tail distribution. Typically, the alternative hypothesis is $H_1 : \mu \neq \mu_0$.



Suppose we are doing a hypothesis test under the significance level α, the region of accepting the H_0 is

$$-z_{\alpha/2} \leq \frac{\bar{X} - \mu}{SE} \leq z_{\alpha/2}$$

such that the rejection region (p-value) has probability α. This is equivalent to

$$\bar{X} - z_{\alpha/2}SE \leq \mu \leq \bar{X} + z_{\alpha/2}SE$$

which is exactly the $100(1-\alpha)\%$ confidence interval of $\bar{X}$. Therefore, for a two-sided test, we have the rule:

- Reject H_0 if μ is not in the $100(1-\alpha)\%$ CI: $\bar{X} \pm z_{\alpha/2}SE$

We conclude this chapter by reiterating a couple of critical points that could be easily misunderstood.

Rejecting H_0 doesn't means we are 100% sure that H_0 is false. We might make Type 1 errors. Setting a significance level just guarantee we won't make Type 1 error too often.

Failing to reject H_0 does not necessarily mean H_0 is true. We could make a type 2 error when failing to reject H_0. Moreover, unlike type 1 error rate is controlled at a low level, type 2 error rate is usually quite high. When we fail to reject H_0, it just means the data are not able to distinguish between H_0 and H_1. That's why we say *fail to reject*. <u>p-value is not the probability that the H_0 is true.</u>

Saying that results are statistically significant just informs the reader that the findings are unlikely due to chance alone. However, it says nothing about the practical importance of the finding. For example, rejecting the H_0: $\mu = \mu_0$ does not tell us how big the difference $|\mu - \mu_0|$ is. Mostly in practice we care more about the magnitude of this difference, rather than the fact that they are indeed different. It is possible that the difference is too small to be relevant even if it is significant.

Epilogue

The furthest distance in the world

Is not between life and death

But when I stand in front of you

Yet you don't know that I love you

— R. Tagore

The greatest distance in the world

Is not between love and hate

But we know each other in this classroom

Yet we become strangers after the semester

The greatest tradegy in the world

Is not life and death

But there are billions of people on earth

But we are still alone

— Dr. Wang

Goodbye!

Good luck with your final exam!